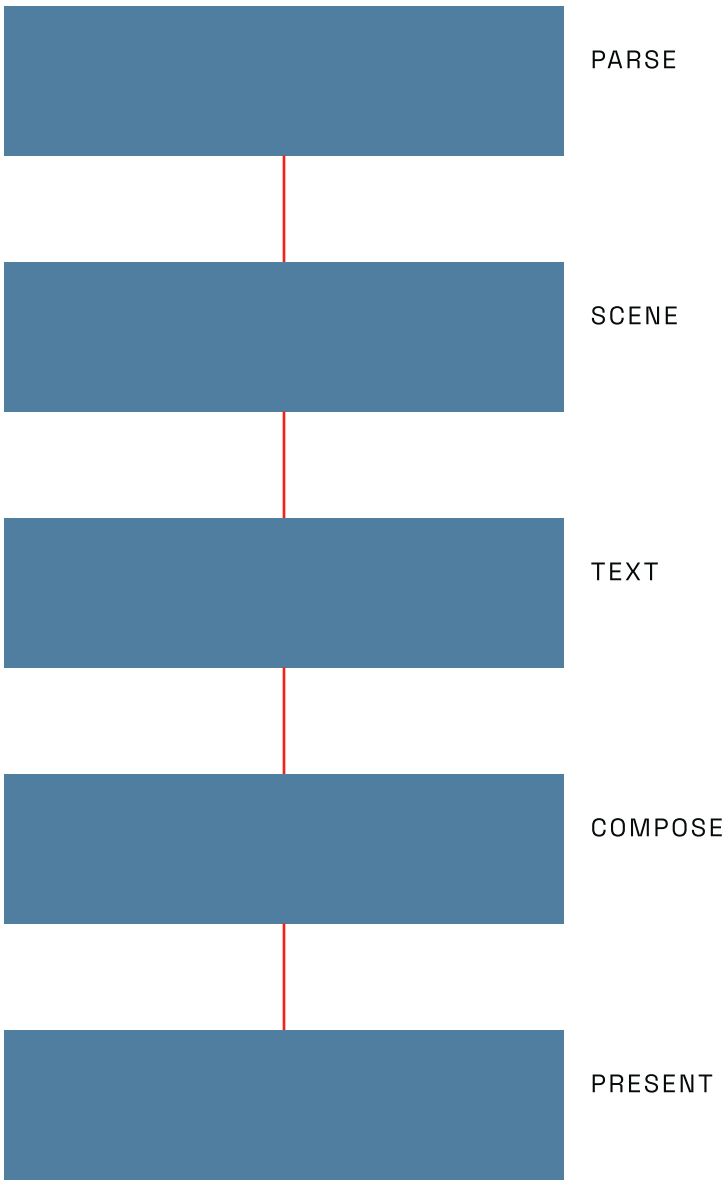
THE PAGED ANNUAL

Volume One — a specimen of the composing engine, set entirely by the engine it describes.


PARSE
SCENE
TEXT
COMPOSE
PRESENT

The Paged Annual

Volume One

A specimen of the composing engine, its plugins, and the interchange they serve.

The Paged Annual, Volume One. Set in Source Serif 4, Fraunces, Space Grotesk, EB Garamond, and JetBrains Mono.

Published by the engine that composed it. First edition, MMXXVI.

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Every specimen in this volume was authored over the mutation wire against protocol 62.

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SPECIMEN NO. 3 · PARAGRAPHTABSTOPS · LEADER: DOT

◪ *No generate-TOC op exists on the wire; the fixture's TOC style is declared, the entries here are set live from the plan. → Appendix A*

Contents, continued

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How to read this book

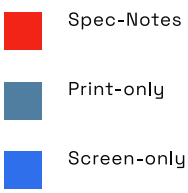
Every demonstration is a numbered Specimen. Its label — set in the outside margin, on the Annotations layer, under the Spec-Notes condition — names the wire op, property path, or registry row it proves. Hide Spec-Notes and the ledger becomes a clean annual; show it and every page cites its own evidence.

■ demonstrated in full · ⊠ demonstrated to a recorded limit, with a margin note pointing at Appendix A · ☒ not modelled by declaration, listed only in the appendix. Nothing on any page pretends.

LAYERS, BOTTOM FIRST

`Grid · Background · Content · Annotations · Notes`

Prose sits on Content; plates on Background; every label on Annotations.



Foreword

A *capability* that cannot be demonstrated in a document is a claim, not a capability. This volume exists to close that distance: every op the wire declares, every property the catalog names, every plugin the platform carries, shown in a page a reader can hold.

The pages that follow are not screenshots and not mockups. They were authored over the same mutation channel the editor's own panels use, against the same engine that renders them, and saved into the same container this file is. Where a limit exists, it is printed. Where a lane is partial, the margin says so.

What survives here is what the system can actually do.

The engine, by op group

TEXT & STORIES 7 ops	FRAMES & PAGES 12 ops	PATH TOPOLOGY 8 ops
PATHFINDER 8 ops	GROUPING 6 ops	LAYERS 7 ops
COLOUR & SWATCHES 13 ops	STYLE CRUD 16 ops	TABLES 13 ops
THREADING 2 ops	SECTIONS & GUIDES 6 ops	MASKS · PATH TEXT 4 ops

The studios

PAGED.DRAW
92 commands

PAGED.IMAGE
128 kernels

PAGED.SHEET
224 functions

PAGED.WEB
7 commands

PAGED.DATA
42-fn DSL

PAGED.DOC
DOCX both ways

PAGED.PDF
PDF in

PAGED.PUBLISH
IDML both ways

Part I — The Document

SPECIMEN NO. 6 · D-PLATE — NO FURNITURE · AUTHORED FIELD + TITLE

A book that sets itself owes its reader an account of how. The next eight pages are that account: the grid this page hangs on, the masters that stamp its furniture, the layers it stacks, and the conditions under which any of it is visible.

1

Anatomy of This Book

In which the volume opens itself on the table: margins, masters, layers, and the apparatus that decides what prints.

Every chapter after this one demonstrates something the engine does to content. This chapter demonstrates what the engine does to the book itself. The pages that follow are their own exhibit: a spread that draws its own grid, a page whose running head is a recorded act of disobedience, a page that is created, reshaped and destroyed without the book ever changing length, and a page that shows the same sentence three ways depending on who is looking.

Nothing here is illustration. The hairlines on the next spread sit at the same coordinates the margin preferences declare; the master tiles on page 16 name the seven masters this file actually carries; the layer bands on page 17 are read from the document, not typed. Where the machinery has a limit, the margin says so and Appendix A holds the ledger.

The grid, shown working

This spread is printed on top of its own construction. The hairline box is the margin preference — 54 over 81, 48 inside, 60 outside, mirrored across the gutter. The six tinted columns are the body grid, 62 points on a 12 point gutter. On the facing page, every thin rule is one 13 point baseline; every leading value in the style battery is a multiple of it.

One ruler guide is live on this spread right now: inserted, moved down thirteen baselines by the wire, and left standing while its vertical companion was deleted. You cannot see it here, and that is the point the margin records.

◪ *Guides are an editor overlay: the wire holds them (SpreadSummary.guides) and the canvas draws them violet, but they are never emitted into the display list or any export. The surviving guide on this spread is real and invisible → Appendix A.*

An override, kept

Look at the top of this page. Every other body recto in this book carries the running head the B-Body master stamps; this one reads SPECIMENS RECONSIDERED · AN OVERRIDE instead. That head is not typed on this page by this chapter — it is a page-local override recorded in the fixture, and the machinery under it is the demonstration: a page that overrides a master item suppresses the master's copy without detaching from the master.

Everything else the master stamps — the folio below, the verso head opposite — keeps tracking the master. Delete the override and the stamped head would return; re-apply the master and the override would still win. One item disagrees; the rest of the furniture does not even notice.

Seven masters, one transient page

A master spread stamps furniture — running heads, folios, rules — onto every page that applies it. This page applies E-Data, which is why the fine twelve-column furniture is here without this chapter drawing it. The seven-master set below is the whole cast of this book.

While this page was being set, an eighth page briefly existed. It was inserted after page 134, given the B-Body master (the render changed: the head and folio appeared on a page nothing had drawn on), reassigned to D-Plate (the same op, to the master that stamps nothing), duplicated, resized to half height, and then deleted along with its twin. The count you can check at the foot of this book never moved.

THE MASTER SET, AS THIS FILE DEFINES IT

A-FRONT front matter	**B-BODY** text spreads	**C-OPENER** chapter openers	**D-PLATE** full-bleed plates
E-DATA 12-column data	**F-VERTICAL** CJK vertical	**G-APPENDIX** apparatus	

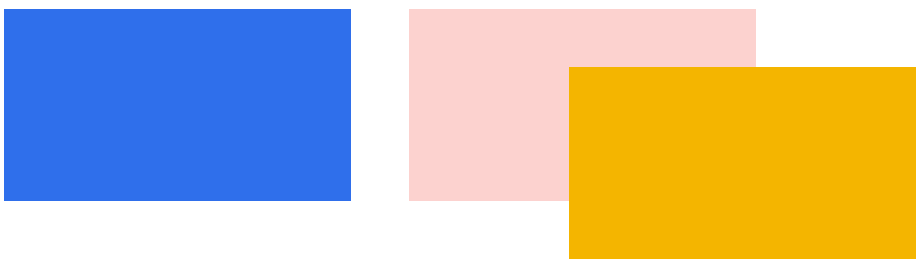
SPECIMEN NO. 10 · INSERTPAGE / DUPLICATEPAGE · RESIZEPAGE / DELETEPAGE · APPLYMASTERTOPAGE · DEMONSTRATED, NOT RESIDENT

◪ *The masterPages collection reports self ids only — label is a copy of the id — so the seven names above were recovered from the document's own IDML export rather than resolved through a collection → Appendix A.*

The layer plan, exercised

A layer is a band of stacking order with three switches: visible, locked, printable. This book keeps five, drawn as bands below, bottom first — the grid under everything, the plates over it, the prose over those, and the apparatus on top. While this page was set, a sixth existed: born as Scratch, renamed to Scratch Bench, handed the blue exhibit below, switched invisible and back (the render changed both ways), moved to the bottom of the stack, and removed.

The removal is worth reading twice. The engine does not refuse to remove an occupied layer: the layer record goes, and the item keeps a reference to a layer that no longer exists until someone re-homes it — which this page did, to Content, where the exhibit now lives. The overlapping pair beside it demonstrates the other half of stacking: Arrange, front and back, with no geometry change at all.



Left: the exhibit that survived its layer. Right: the marigold plate was sent to back, then brought to front — reorderElement, both directions.

THE STANDING BANDS, BOTTOM FIRST

Grid

Background

Content

Annotations

Notes

SPECIMEN NO. 11 · LAYERINSERT / SETNAME / MOVE · SETVISIBLE / SETLOCKED / SETPRINTABLE · LAYERREMOVE: ALLOWED OCCUPIED · ITEMLAYER · REORDERELEMENT

Conditions and variables

This sentence is bound for the press run and nowhere else.

This sentence exists only where a screen renders it.

This sentence is production apparatus, visible while the working copy shows its notes.

Line one carries Print-only, line two Screen-only, line three Spec-Notes. While this page was set, Screen-only was switched off and the second line left the render; switched back on, it returned.

TEXT VARIABLES, READ FROM THE DOCUMENT

```
Edition — CustomTextType — resolves to "Edition: MMXXVI"
Running Header — RunningHeaderType — picks up FirstOnPage of its named style
Chapter — ChapterNumberType — resolves at composition time
Page Count — PageCountType — resolves at composition time
```

A variable is a slot the composer fills, and the running heads of this book are the working demonstration. The verso head is plain master text — THE PAGED ANNUAL · MMXXVI on every body verso. The recto head is a Running Header variable: after layout, it resolves to the first Chapter Title paragraph at or before its page, which is why the rectos of this chapter carry the title set on page 13 without any page carrying that text itself.

Custom-text and page-count variables resolve from the designmap at emit; date variables format a deterministic clock. Every one of them keeps a baked ResultText as its fallback, and the margin records where the fallback is what you would see.

Where no Chapter Title paragraph precedes a page, the Running Header variable prints its baked ResultText — the literal word Chapter in this fixture. The registry row still records renderer support for header/chapter variables as partial → Appendix A.

2

The Letter

A type specimen does not prove that a face is beautiful. It proves that the engine can ask the face the right questions.

The faces in this book carry real feature tables: EB Garamond answers for swashes, small capitals, old-style figures and seven stylistic sets; JetBrains Mono for the slashed zero; Space Grotesk for tabular figures. The next spread asks for each feature by its property path and shows both states side by side.

The terms are strict. A row whose feature the composer actually drives — ligatures, case, position, and every metric transform on the closing page — shows a genuine difference. A row whose feature the wire can only record is printed anyway, twice and identically, with the margin naming the limit. A specimen that hid its second kind of row would be an advertisement.

The feature specimen

liga · characterLigatures — left ON (the default), right OFF

fine offices affirm floral scripts | fine offices affirm floral scripts

A

dlig · characterOtfFeatures="dlig" — authoring metadata; both halves print identically

distinct strategist acts | distinct strategist acts

characterCase · plain | Small Caps style | SmallCaps path | AllCaps path on lowercase source

Letterforms | Letterforms | Letterforms | lETTER-RHYTHM

figures · OTFFigureStyle rides these paragraph styles' definitions; run resolution never reads it — and onum via characterOtfFeatures is metadata

0123456789 · 31,556,952 — Annual Body, ProportionalOldStyle

0123456789 · 31,556,952 — Table Number, TabularLining

frac · characterOtfFeatures="frac" · JetBrains Mono — authoring metadata

1/2 3/4 7/8 22/7 | 1/2 3/4 7/8 22/7

◪ *characterOtfFeatures stores a tag list the shaper never reads; the raster obeys only the discrete OTF toggles of runs and styles, which no wire door writes. Rows so marked print both states alike → Appendix A.*

The feature specimen, continued

ordn · plain | characterOtfFeatures="ordn" (metadata) | characterPosition="OTSuperscript" on the suffixes (rendered, metric)

1st 2nd 3rd 4th | 1st 2nd 3rd 4th | $1s^{t}$ $2n^{d}$ $3r^{d}$ $4t^{h}$

characterPosition · Superscript and Subscript, position-metric size and shift

E = mc2 | $H2_{O}$ | $Fe3^{+}$ | log2 n

swsh · EB Garamond Italic · characterOtfFeatures="swsh" — authoring metadata

Rare Kingdoms Answer | *Rare Kingdoms Answer*

ss01 · characterOtfFeatures="ss01" — authoring metadata

Quirky Regard gently | Quirky Regard gently

zero · JetBrains Mono (its zero is slashed by design) · characterOtfFeatures="zero" — authoring metadata

0 100 2026 0x0F | 0 100 2026 0x0F

hist · characterOtfFeatures="hist" — metadata; the discrete OTF lane carries no hist flag at all

his historic listens | his historic listens

◪ *The GSUB rows here demonstrate the authoring surface, not the substitution: the tag list persists, exports and reads back unchanged. The ordinal and position rows are metric transforms the composer does perform → Appendix A.*

Metal that moves

characterTracking · thousandths of an em, five stops

letterforms breathe (-50)
letterforms breathe (+0)
letterforms breathe (+50)
letterforms breathe (+120)
letterforms breathe (+200)

characterHorizontalScale 70 / 130 · characterVerticalScale 160

condensed to 70
expanded to 130
tall at 160

characterSkew 12 · characterBaselineShift +4 / -4

a *false italic, twelve degrees of skew*
steady high steady low steady

characterKerningMethod · the pair fitting the composer applies

WAVE To AVATAR Yield — Metrics
WAVE To AVATAR Yield — Optical
WAVE To AVATAR Yield — None

characterUnderline · characterStrikethru — paint-only, no reflow

underlined for the record
~~struck from the record~~

SPECIMEN NO. 15 · CHARACTERTRACKING / SKEW · CHARACTERH/VSCALE · CHARACTERBASELINESHIFT · CHARACTERKERNINGMETHOD · CHARACTERUNDERLINE / STRIKETHRU

◪ *Optical kerning falls back to metrics — the middle kerning sample and the first render identically by design of the current composer → Appendix A.*

3

The Paragraph

Letters make words; the paragraph is where they start to argue. A composer worth the name does not break lines one at a time - it weighs the whole paragraph at once. This chapter watches it do so: the breaks, the keeps, the caps, and the parameters that steer them.

One paragraph, two disciplines

The paragraph to the right and the paragraph facing it across the gutter are the same text in the same measure. They differ by two range properties and nothing else.

On this page: paragraphHyphenation true, paragraphJustification LeftJustified. The composer may divide words at the hyphenation points the TeX patterns license, and it flexes the spaces of every line between the H&J minimum and maximum so the right edge lands flush. Opposite: paragraphHyphenation false, paragraphJustification LeftAlign. No division, no flex - each break is simply the last word that fits.

Below, the parameter row: the same passage three times under characterTracking minus twenty, zero, and plus sixty thousandths of an em. Spacing pressure moves the breaks.

A paragraph is not broken one line at a time. The composer reads the whole of it first, weighs every place a line could end, and charges each candidate for the space it stretches, the word it divides, and the shape it leaves behind; then it chooses the cheapest complete set of breaks. This is the total-fit method Knuth and Plass described in 1981, and it is why a justified column can hold an even colour from its first line to its last. Forbid the hyphen and refuse the stretch, and the same sentences must end wherever the words happen to stop: the right edge frays, and the ragged column tells you, plainly, what the algorithm had been doing for you all along.

paragraphJustification = LeftJustified · paragraphHyphenation = true

The measure is narrow on purpose: at this width every extra unit of tracking costs a word to the next line, and the breaks move.

characterTracking = -20

The measure is narrow on purpose: at this width every extra unit of tracking costs a word to the next line, and the breaks move.

characterTracking = +0

The measure is narrow on purpose: at this width every extra unit of tracking costs a word to the next line, and the breaks move.

characterTracking = +60

A paragraph is not broken one line at a time. The composer reads the whole of it first, weighs every place a line could end, and charges each candidate for the space it stretches, the word it divides, and the shape it leaves behind; then it chooses the cheapest complete set of breaks. This is the total-fit method Knuth and Plass described in 1981, and it is why a justified column can hold an even colour from its first line to its last. Forbid the hyphen and refuse the stretch, and the same sentences must end wherever the words happen to stop: the right edge frays, and the ragged column tells you, plainly, what the algorithm had been doing for you all along.

paragraphJustification = LeftAlign · paragraphHyphenation = false

READING THE COMPARISON

Count the hyphens in the justified column, then look for the lines they rescue: without them the composer would be forced to stretch a three-word line across the full measure. The ragged column pays the other way - no line is strained, but the silhouette gives up its even edge.

The flex itself is governed by the H&J limits - minimum, desired and maximum word spacing, letter spacing, and glyph scaling. The engine parses and honours all three families when it composes the justified column above; none of them is yet a property path on the wire, so the parameter row beneath uses tracking, the spacing control that is.

The fixture's body style also declares a 36 pt hyphenation zone. The zone steers ragged hyphenation only - for justified text the engine zeroes it, deliberately, to mirror InDesign - and it too has no wire path, so the comparison you see is the style's declaration at work, not a live write.

◪ *H&J spacing limits and the hyphenation zone parse and render but have no property path; shown via style declarations + tracking. → Appendix A*

Keeps and breaks

A short lead paragraph occupies the top of the measure.

This paragraph was set at the foot of its column on purpose. Where it may break freely, the column boundary cuts it mid-thought and strands its final lines in the next frame. With keep-lines-together set, the composer is asked to move the whole paragraph across the boundary instead, trading an early gap in the first column for an unbroken argument in the second. The flag rides the wire either way; the pixels report what the composer does with it.

paragraphKeepLinesTogether = false · the boundary cuts the paragraph where the first column ends

A short lead paragraph occupies the top of the measure.

This paragraph was set at the foot of its column on purpose. Where it may break freely, the column boundary cuts it mid-thought and strands its final lines in the next frame. With keep-lines-together set, the composer is asked to move the whole paragraph across the boundary instead, trading an early gap in the first column for an unbroken argument in the second. The flag rides the wire either way; the pixels report what the composer does with it.

paragraphKeepLinesTogether = true · the same text, asked to cross the boundary whole

Body text fills this column so that the heading below arrives at its very foot, the classic place for a heading to be orphaned.

A HEADING THAT MUST STAY WITH ITS TEXT

The paragraph the heading announces. Keep-with-next asks the composer to carry the heading over to this side of the boundary rather than leave it stranded at the foot of the first column.

paragraphKeepWithNext = 1 on the heading

DROP CAPS announce a beginning. The first character of this paragraph is set three lines deep, and the opening words carry the Lead-In character style in small capitals, the classic pairing of a cap and its runway.

Space before, 13 pt, opens this line.
Space after, 13 pt, follows this line.

First-line indent, 26 pt: only the opening line of this paragraph steps in.

Left and right indents, 18 pt each, narrow this whole paragraph from both sides.

4

The Style

Nothing in this book is formatted by hand. Every paragraph wears a named style, and every style stands in a chain of inheritance that ends at Annual Body. This chapter turns the system on itself: the cascade, live creation and deletion, nested styles, numbered lists, and the rules and tab stops a paragraph can carry.



The cascade, printed by itself

A style rarely says everything it means. Body First declares only that its first line does not indent; its font, size, leading, ink and justification all arrive from Annual Body through the basedOn chain, resolved at compose time. Each line below prints its own chain, set in its own style - the specimen and the citation are the same ink. The chains are read live from the paragraphStyles collection, not typed, so this page cannot drift from the document it sits in.

Annual Body - the root; every chain below ends here.

Body First <- Annual Body

Body Small <- Annual Body

Footnote <- Annual Body

- Bullet List <- Annual Body

1. Numbered 1 <- Annual Body
2. Numbered 2 <- Numbered 1 <- Annual Body

Catalog Entry ← Annual Body

Index Sub <- Index Entry

Caption - declared free-standing, no basedOn

Margin Note - declared free-standing, no basedOn

SPEC LABEL - DECLARED FREE-STANDING, NO BASEDON

Read the list as evidence, not decoration: Numbered 2 walks two hops to the root and inherits its numbering through the first; Caption stands alone and must say everything itself. Direct formatting still outranks any of them - the cascade resolves direct overrides above character styles above paragraph styles - which is how the next page will dress a style it has only just invented.

Live style surgery

This page performed surgery on its own wardrobe. The style these words wear, Field Note, did not exist when the chapter began: createParagraphStyle minted it, based on Annual Body, and eight setStyleProperty writes dressed the definition - 8.5 pt type, slate ink, a little tracking, half-rhythm space before and after, no first-line indent, flush left, and a next-style pointing back at Annual Body.

It was born under the working name Field Memo and renamed in front of you; renameParagraphStyle changes only the display name, so the reference every one of these paragraphs holds survived the change. The letterspaced vermilion openers are Field Note Caps, a character style created the same way and applied as a range on each paragraph.

A second pair, Scratch Note and Scratch Caps, went through the whole triple - created, renamed, deleted - in the same breath. The deletions are real wire ops, so those styles are demonstrated, not resident: the checkpoint this chapter saves carries no trace of them.

The vermilion rule under each paragraph is a range property, not part of the definition. The style-definition door carries eight paths today; font family, leading and rules are not among them - the first two arrive here through the basedOn chain, the rule through the range door.

Field Note, created and dressed on this page · setStyleProperty: characterFontSize, characterTracking, characterFillColor, paragraphSpaceBefore, paragraphSpaceAfter, paragraphFirstLineIndent, paragraphJustification, paragraphStyleNextStyle

SPECIMEN NO. 32 · CREATEPARAGRAPHSTYLE · RENAMEPARAGRAPHSTYLE · DELETEPARAGRAPHSTYLE · CREATECHARACTERSTYLE · RENAMECHARACTERSTYLE · DELETECHARACTERSTYLE · SETSTYLEPROPERTY X11

◪ *The style-definition door carries 8 paths; font, leading and rules are basedOn or range matters. Scratch styles deleted.* → *Appendix A*

Nested styles and the next style

NO-114. Field notebook, gridded, ninety leaves - 4.50
NO-201. Composing stick, brass, adjustable - 18.00
NO-350. Pica ruler, steel, twelve inch - 6.25
NO-512. Type-high gauge, calibrated - 11.40

No range op touched the stock numbers above. Catalog Entry carries a nested style in its definition: through one occurrence of the delimiter, a period, inclusive, the run wears Specimen Number - the mono slate hand you see on each SKU. Apply the paragraph style and the restyling happens on its own, at compose time, wherever the delimiter lands.

The model names twelve delimiter kinds a nested style can count: characters, words, sentences, any digit, any letter, double quotes, single quotes, a tab, a forced line break, the end-nested-style marker, a literal character, and an unknown fallback that fires at the start. This page demonstrates the literal character; the front matter's tab-led contents would suit the tab kind just as well.

THE NEXT STYLE, AS DECLARED

```
Chapter Title → Deck → Body First → Annual Body
Field Note → Annual Body
```

Both chains above are read live from the paragraphStyles collection - the first is the fixture's opener chain, the second was forged two pages ago when setStyleProperty gave Field Note a next-style of Annual Body. NextStyle answers one question: when a writer ends a paragraph at the keyboard, what style should the next one wear? A chapter title hands off to its deck, the deck to the first body paragraph, and the writer never touches a style menu. This book, poured over the wire, applies every style explicitly - which is why the chain is printed here as evidence rather than performed.

◪ *NextStyle fires on interactive typing, not on wire pours; the chains are shown as live-read data. → Appendix A*

Bullets and numbering

- A bulleted paragraph is a style choice, not typed glyphs.
- The marker comes from the definition: the default round bullet.
- Delete the style and the bullets would vanish with it.

- These two lines were plain body text until two range writes made them a list.
- Their dash marker is paragraphBulletCharacter, set live.

Bullet List (style) · paragraphListType = BulletList + paragraphBulletCharacter (live)

Three ops govern the list resources. createNumberingList minted Annual Steps, the list both numbered frames opposite are bound to through paragraphAppliedNumberingList. It was born story-scoped, so the lower frame - a separate story - restarted at one; editNumberingList then set ContinueNumbersAcrossStories and the lower frame carries on from the upper. A third list, Scratch List, was created and deleted again: demonstrated, not resident.

The lettered entries restart because they live in their own story. An explicit start-at is not a property path on the wire, so a sub-level restarts by story scope here; the margin note records the boundary.

1. Mix the ink.
2. Set the line.
3. Pull a proof.

    a. Check the margins.
    b. Check the colour.

Numbered 1 + appliedNumberingList = Annual Steps · sub-level Numbered 2, paragraphNumberingFormat = a, b, c

4. Wash the form.
5. Distribute the type.
6. Note the run.

The same list in a second story: born restarting at 1, it continues at 4 since editNumberingList set ContinueNumbersAcrossStories

SPECIMEN NO. 34 · CREATENUMBERINGLIST · EDITNUMBERINGLIST · DELETENUMBERINGLIST · APPLIEDNUMBERINGLIST · PARAGRAPHLISTTYPE · PARAGRAPHBULLETCHARACTER · PARAGRAPHNUMBERINGFORMAT

NumberingStartAt and per-level restarts are not on the wire; the lettered sub-list restarts by story scope. → *Appendix A*

Rules, shading, borders, tabs

Ruled matter: this paragraph carries a vermilion rule above and a slate rule below, each a whole ParagraphRule struct - on, colour, weight, offset - set on the range in a single write.

paragraphRuleAbove: Vermilion, 1 pt · paragraphRuleBelow: Slate, 0.5 pt

Left stop, 132 pt..... flush left after the leader
Centre stop, 240 pt..... centred on the stop
Right stop, 420 pt.....flush right
Decimal stop, 300 pt.....1,204.75

paragraphTabStops · LeftAlign / CenterAlign / RightAlign / CharacterAlign with alignmentCharacter = period, leader = dot

ink	12.5
paper	1,204.75
thread	0.9

One CharacterAlign stop shared by three lines: the points sit plumb

A paragraph rule is not an underline: it belongs to the paragraph, spans the measure less its indents, and rides the wire as one struct, so a single write replaces colour, weight and offset together. Tab stops travel the same way - the whole TabList replaced per write, which is why each menu line above carries its own private stop.

Paragraph shading and paragraph borders are the missing guests at this table: the engine parses both from IDML and renders both faithfully, but neither has a property path, so a live page cannot yet author them. They appear in this book only where a fixture declared them - and honestly, in the margin, where they do not.

◪ *Paragraph shading and border parse + render but have no property path on the wire; not authorable live, not faked. → Appendix A*

5

The Story

A paragraph belongs to a story, and a story is not a frame: it is the text itself, which merely stands in frames for as long as they hold it. This chapter threads one story through four frames, lets another overflow and says so, and teaches frames to balance their columns, grow to fit, wrap around an island, and carry anchored cargo through every edit.

One story, four frames

The flagship demonstration of this chapter: the paragraphs below were poured once and flow through four linked frames — two on this page, two on the facing one — one address space of text wearing four coats.

B

One story, in four frames.

The frame is furniture.

The story is the text itself.

It was poured exactly once.

It landed in the first frame.

That frame filled, line by line.

Composition carried on unbroken.

The second frame took the rest.

Then it crossed the gutter.

A third frame stood waiting.

The facing page received it.

Nothing here was copied.

Nothing here was retyped.

Delete a sentence above this one.

Every line below climbs one stand.

Widen any frame in the chain.

The others give up their lines.

Or they take on more of them.

The chain is a single sequence.

Four coats, one body of text.

The proof is not the look of it.

Each frame was asked directly.

The hit-test names a story id.

All four answer with one id.

That answer is the thread.

This is how a feature crosses pages.

A magazine lives on such chains.

A chain is not a copy of frames.

SPECIMEN NO. 51 · LINKFRAMES ×3 · ONE POUR · ORACLE: STORYOF PER FRAME · STORYCHARS = POURED CHARS

◪ *A paragraph that crosses a frame boundary carries one line into the next frame and scatters further overflow one line per frame — so this chain is set in one-line paragraphs and composes at paragraph granularity.* → *Appendix A*

A footnote is the page apologising for an interruption it refuses to omit. The engine reserves its space through a compose, measure, and re-compose fixpoint that vertical justification then respects; the rule above the pool comes from the document footnote options, authored in the base fixture because no mutation writes them.12

The reservation fixpoint: compose, measure the pool, compose again.

An oversized footnote does not yet split across frames — a recorded limit.

Above: the fixture's footnote exhibit — a body frame whose story anchors two live footnotes. The engine reserves their space through a compose, measure, re-compose fixpoint; no mutation on the wire writes a footnote, so the exhibit ships in the base document.

It is one address space of text.
The wire calls the join linkFrames.
The story owns the walk order.
Unthread the chain and it stops.
Each frame keeps what it holds.
Thread it again: one sequence.

Unbroken, unretyped, in order.
Set wide or set narrow at will.
The story neither grows nor loses.
Not a letter appears twice.
What you read is the engine.
It is reading itself back.
In order, frame after frame.
It ends low on the recto.
With room to spare, as promised.
One story, many stands.

SPECIMEN NO. 52 · FOOTNOTE EXHIBIT (FIXTURE) · UNLINKFRAMES + RELINK · DEMONSTRATED, NOT RESIDENT

An oversized footnote does not split across frames; footnote bodies set in an 8 pt MVP under the fixture's separator rule. → Appendix A

Overset, honestly

The tinted frame below was poured more text than it can show. Overset text is editor state: retained by the model, saved by the container, clipped by the render, and reported by the engine as a diagnostic — the assertion under this page read that diagnostic, not the pixels, before the frame was threaded on.

An overset is not a loss; it is a fact about fit, recorded rather than hidden. The story you are reading was poured, whole, into a frame two sizes too small for it, and the engine did what an honest compositor must: it set what fits, clipped the remainder from the render, and reported the difference as a diagnostic instead of quietly discarding it. Every character of this paragraph is still in the model —

Continued on page 26 — said every jump line ever printed; this one is set on page 26.

the container saves it, the exports carry it, and the stories read-out counts every one of them — but the page shows only what the frame can hold. This is editor state, not page state: give the frame another hundred points of depth, or thread it onward to a second stand, and the hidden text returns without being retyped, because it never left. The red badge a designer sees on an overset frame is this same diagnostic wearing its working clothes. This page did the second: the overset was read while the tinted frame stood alone, the frame was then threaded to the stand below, and the rest of the paragraph came back on its own.

SPECIMEN NO. 53 · STORIES: OVERSET FLAG, THEN LINKFRAMES · INSERTFIELD: PAGENUMBER · INSERTFIELD: NEXTPAGENUMBER

setFieldValue does not drive these two markers — they resolve at compose time; the op exists for plugin placeholder fields. → Appendix A

Columns, balance, depth, and the first baseline

Three frames, three text-frame preferences: a two-column split with a ten-point gutter, the same split balanced so the columns end even, and a vertically justified box whose inset holds the text off its own stroke. Below them, the six first-baseline policies in identical boxes.

A column is a promise about line length: split the box in two and every line stays short enough to read without effort. The gutter is the fence between them — ten points of it here.

Two columns, 10 pt gutter.

Balancing splits the text so both columns of this box end within a line of each other.

Balanced — both columns end even.

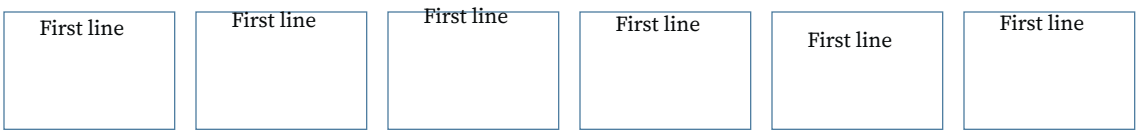
The top of the box.

The middle holds its ground.

The bottom of the box.

Vertical justify, 10 pt inset off the stroke.

FirstBaselineOffset — the same two words in the same 44 pt box, six ways:



AscentOffset · CapHeight · XHeight · EmBoxHeight · LeadingOffset · FixedHeight

The box that answers to the text

The same paragraph poured into five boxes, and five auto-sizing modes: the control is authored to fit and stays put; the rest start too small and grow in the directions their mode allows. The page's own proof is a render taken before and after the five property writes — only the growth lies between.

Off — the control: authored to fit; it stays put.

Auto-sizing lets the box answer to the text instead of the text to the box: pour more and the frame grows in the directions its mode allows, holding whichever edges the mode pins in place.

HeightOnly — the bottom edge gives way.

Auto-sizing lets the box answer to the text instead of the text to the box: pour more and the frame grows in the directions its mode allows, holding whichever edges the mode pins in place.

HeightAndWidth — both edges give.

Auto-sizing lets the box answer to the text instead of the text to the box: pour more and the frame grows in the directions its mode allows, holding whichever edges the mode pins in place.

WidthOnly — the right edge gives way.

Auto-sizing lets the box answer to the text instead of the text to the box: pour more and the frame grows in the directions its mode allows, holding whichever edges the mode pins in place.

HeightAndWidthProportionally — the box keeps its shape.

Auto-sizing lets the box answer to the text instead of the text to the box: pour more and the frame grows in the directions its mode allows, holding whichever edges the mode pins in place.

SPECIMEN NO. 55 · TEXTFRAMEAUTOSIZING ×5 · MODE-ONLY; ANCHOR: SEE MARGIN · ORACLE: RENDER DELTA

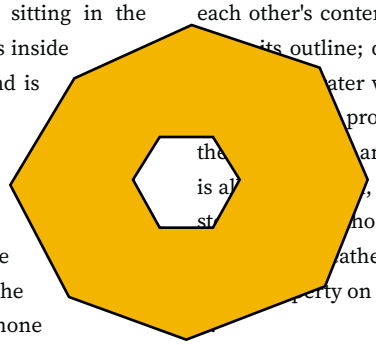
The wire sets the auto-sizing MODE only: AutoSizingReferencePoint — the anchor the growth holds — is parse- and render-side, with no catalog path to write it, so every box here grows from the parser-default anchor. → Appendix A

The wrap catalog

Text wrap is the page's negotiation between type and object: the object declares an exclusion, and every line that would again have crossed it is measured against what remains. The mode is declared. A bounding-box wrap excludes the object's rectangle and nothing more; a contour wrap traces the outline itself, so the text follows every bay and headland of the island sitting in the middle of this measure — and because its inside edges are open, a hole through the island is open water too.

A jump-object wrap concedes the whole band: no line shares a baseline with the object, however narrow it is. A next-column wrap concedes the rest of the column, sending the interrupted line to the top of the following one. And a wrap of none concedes nothing — the text runs beneath the tint as if it were not there, which is why the fourth tile reads through its own veil. Each declaration here was applied as a property, in one batch, after this prose was already set: the reflow you a batch landing.

Every wrap on this page is symmetrical about one fact: the pushed text and the pushing object know nothing of each other's content. The columns re-measure; the island its outline; delete the island and the measure clos- ater without a seam. The inverse case at the proves the rule by turning it around — there th an exclusion but the only territory the text is a , and everything outside it is given up in- st hose two poles sit the four tiles above: each ather system in the flow, each declared by a rty on a single object, each undone by clearing



Clockwise from top left: BoundingBox pushes the measure off the tint's rectangle; JumpObject clears the full band; the marigold island wraps to its own contour, hole open; None lets the text run beneath the veil; NextColumn sends the interrupted line onward. Below: the inverse wrap, text held inside the outline.

The inverse wrap keeps this paragraph inside the tinted plate not away from it.

SPECIMEN NO. 56 · FRAMETEXTWRAPMODE ×6 · CONTOURTYPE: SAMEASCLIPPING · TEXTWRAPINVERT · INCLUDEINSIDE · FRAMEPATH: 2 SUBPATHS (HOLE)

◪ *A wrap contour excludes EVERY intersecting text frame — there is no per-frame ignore-text-wrap door — so the classic pull quote seated in its own wrap boundary is inexpressible; wrapped objects here are shapes and panels. → Appendix A*

The anchored object

An anchored object is a page item that has surrendered its independence: it belongs to a character in the story, travels with that character through every edit, and reflows the line that carries it. An anchored rectangle entered this very sentence just here over the wire — and paints nothing, contributes no advance, and shifts no line: the sweep's recorded finding, met again live. The other seats — above-line bands, custom positions hung from reference points and offsets, spine-relative plates that flip with the fold — exist in the model and render from parsed documents, but on this wire they cannot yet be reached; the margin carries the exact refusals.

The anchor above is real and paints nothing. Insert #2, measured this run: minted its own id (the duplicate-self_id collision this page used to record was fixed in core v0.63.0). Property battery, measured this run: all ten anchored* paths applied cleanly on this run. Across runs the same battery has both applied and refused node-not-found — the flakiness is the finding.

SPECIMEN NO. 58 · INSERTANCHOREDFRAME (PAINTS NOTHING) · SECOND INSERT: OUTCOME MEASURED PER RUN · ANCHORED* BATTERY: OUTCOME MEASURED PER RUN

The wire-minted anchored frame paints nothing, and property writes on it land in some runs and not in others. Anchored rendering is proven from parsed documents. → Appendix A

6

Scripts of the World

Latin is one script among many, and a composing engine that stops at it has stopped early. In this chapter the text runs the other way — Arabic and Hebrew from right to left, with digits and Latin nested inside — and then turns on its side, in the vertical Japanese the fixture carries. Where a lane is partial, the margin says exactly how.

Right to left

A bidirectional line is stored in logical order — the order it is spoken — and painted in visual order. The renderer reorders the runs at layout: Arabic and Hebrew read right to left, while digits and Latin words nested inside them keep their own left-to-right direction.

الكتابة العربية تجرى من اليمين إلى اليسار.

Arabic — Noto Sans Arabic, right-aligned. The glyphs join and the line runs from the right margin.

כתיבה עברית רצה מימין לשמאל.

Hebrew — Open Sans, right-aligned. The shipped face carries the Hebrew block, and the fonts read-out reports the family resolved.

يمزج هذا السطر العربية مع الكلمة paged 2.0 والأرقام 123 فى اتجاهين.

Mixed direction — the word paged, a version number and the digits 123 ride inside the Arabic sentence, each keeping left-to-right order while the line around them runs right to left.

Two scripts, two shipped faces: Noto Sans Arabic for the Arabic and the mixed line, Open Sans for the Hebrew. Reordering is the renderer's; the faces only supply the glyphs. → Appendix A

縦書きは文字を上から下へ組み、行は右から左へ進む。漢字強調点。

Right: the fixture's vertical exhibit. Columns run top to bottom, lines advance right to left; the face is Noto Sans JP at thirteen points.

No wire op writes StoryDirection — the two vertical stories ship in the base document; everything around them is set live.

Inside the column: the second run carries a GroupRuby reading over its base characters; the final run carries kenten emphasis marks beside each glyph.

Vertical composition turns the writing frame on its side: glyphs stack down the column and columns advance toward the spine. Kinsoku keeps forbidden characters off line starts and ends; mojikumi tightens the full-width punctuation between them.

◪ *Kinsoku is enforced as a hard rule keyed on the type's presence — the push-in/push-out flavours are deferred; mojikumi is a uniform half-width tightening, not the per-adjacency table. → Appendix A*

Right: the same vertical story form on the verso — the columns advance toward the spine on both pages of the spread.

縦書きは文字を上から下へ組み、行は右から左へ進む。漢字強調点。

Ruby sets a reading above its base text — in vertical writing, to the right of the column. Kenten sets an emphasis mark beside each glyph it covers; both ride the run, not the frame, and reflow with the story.

◪ *Ruby renders as a centred group reading: a PerCharacter request collapses to the group form. Kenten renders a filled circle whatever kind is asked for — this fixture asks for sesame dots.* → *Appendix A*

The apparatus of the annual begins here. See the reconsidered specimens overleaf, or visit paged.media for the living registry.

Exhibit, fixture-authored: this story was born navigable - a hyperlink text source on paged.media, the cross-reference source that resolves overleaf, and two index markers filed under Typography and Baskerville.

7 The Apparatus

A book is not only its pages; it is the machinery a reader moves by. Links, bookmarks, references and index marks - shown live where a door exists, named honestly where none does.

The exhibit above was born linked; the two spans below were not. Each became a link while this page was set, one insertHyperlink call apiece.

Consult the living registry for the current capability record.

Or turn to the first chapter opener and compare this furniture with its plainer ancestors.

Hyperlinks, read live: 5 — registry-url, back-to-ch01, xref-overleaf, plus 2 authored on this page moments ago.

Bookmarks, read live: 3 — Apparatus, Chapter One, Data Tables. No wire op minted these; the fixture did.

◪ *insertHyperlink is the apparatus's one live door and it mints URL destinations only — the page-flavoured link above is a page:// URL by convention, not a DestinationPage; true page and text-anchor destinations exist here only fixture-authored → Appendix A.*

The destination and the marks

Specimen notes, reconsidered. Jump back to the first chapter to compare settings.

Exhibit, fixture-authored: the receiving end. This story carries the text anchor the p45 cross-reference resolves to, a link back to the first chapter, and a Colophon index marker.

An index is apparatus that accumulates. Six slim tabs ride the fore-edge of verso pages through this book - a to f, on pages 20, 34, 48, 66, 78 and 96 - and each tab's story carries one or two PageReference markers. A marker files its page under a topic; the topic collects pages from wherever its markers landed.

Ten topics are defined at document level, from Typography to Colophon, and the markers scattered here and on the exhibit pages feed them. Appendix A resolves the whole apparatus into the printed index: topic by topic, with the page lists the markers earned.

THE APPARATUS, READ LIVE

Index topics: 10 — Typography, Baskerville, Grids, Spot colour, Lab colour, Footnotes, Data tables, Vertical writing, Appendices, Colophon.

Cross-references: 1 source marker in the document's own export, carried by the hyperlink xref-overleaf (source on the facing page, destination in the exhibit above). The wire's crossReferences collection lists 0.

◪ *Bookmarks, cross-references and index markers are fixture-authored; only hyperlinks are live-authorable. The crossReferences collection answers empty while the xref above exists. TOC resolution is not shown here → Appendix A.*

8 The Object

Everything on a page is an object with a stacking order, a transform, corners, a stroke and, sometimes, other objects inside it. This chapter takes one plain rectangle and asks it every question the wire knows how to ask.

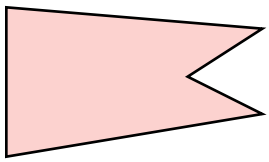
Below, five plates were inserted back to front, A first and E last, then rearranged live: A to the front, B to the front over it, A one step forward and one step backward, E to the back, and finally D to absolute index zero. Every overlap you can read is the paint order saying so.



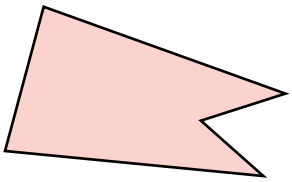
A vermilion, B marigold, C screen blue, D slate, E vermilion 20%. reorderElement walked all five targets - front, back, forward, backward, absolute index - six steps, six gated repaints. Final order, back to front: D, E, C, A, B.

Nine frames, one knob each

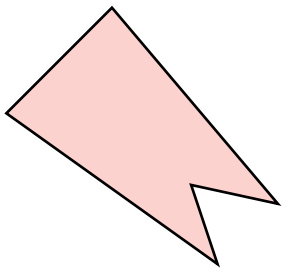
The same pennant nine times. One cell is untouched; seven carry a single transform property write apiece; the last carries the replace-pair - two frames, two different frameTransform histories, one silhouette.



control - no transform

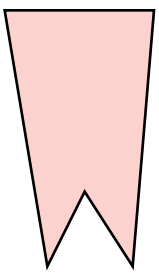


frameRotationAngle 15

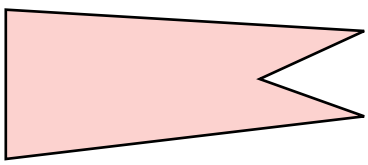


frameRotationAngle 45

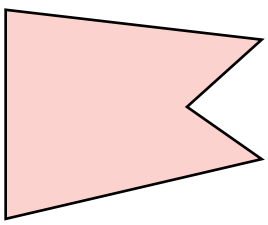
C



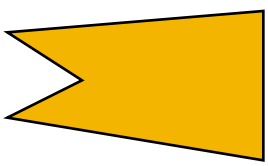
frameRotationAngle 90



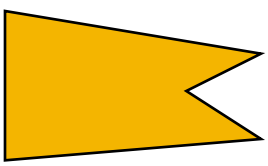
frameScaleX 140%



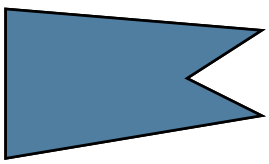
frameScaleY 140%



frameFlipH - notch swaps sides



frameFlipV - slopes trade places



frameTransform x2 vs x1 - one silhouette

The last cell holds two pennants. The vermilion one was moved up 26 points and then given a second frameTransform; the slate one was given only the second. They landed on the same spot - read back, the vermilion frame carries exactly the second matrix and no memory of the first. frameTransform replaces; it never composes.

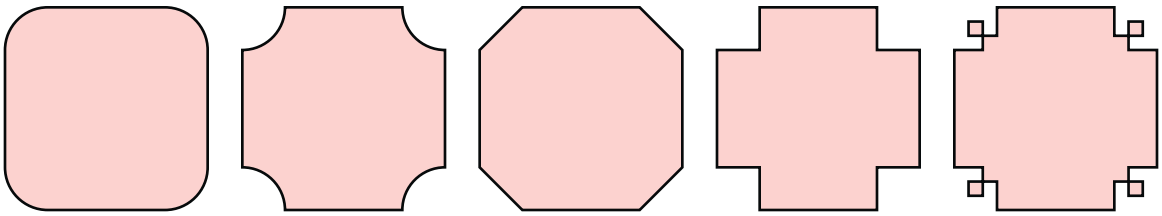
Every rotated, scaled and flipped tile above pre-seeds a translation before its one knob write, because the knob pivots about the item-space origin. The seed is read from the element's own stored geometry, and it is the same arithmetic a rotate-about-centre control performs.

SPECIMEN NO. 66 · FRAMEROTATIONANGLE 15/45/90 · FRAMESCALEX/Y · FRAMEFLIPH/V · FRAMETRANSFORM (REPLACE, PROVED)

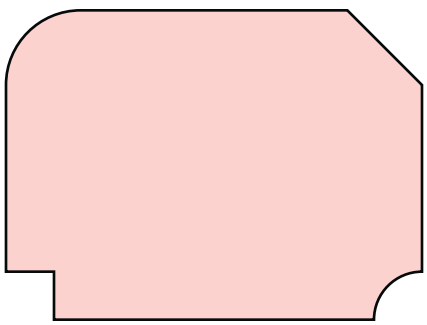
◪ *frameTransform REPLACES the item transform (proved by the pair above); the single-knob writes pivot about the item-space origin with translation preserved, so each tile pre-seeds t = c - Mc to hold its centre → Appendix A.*

The corner sampler

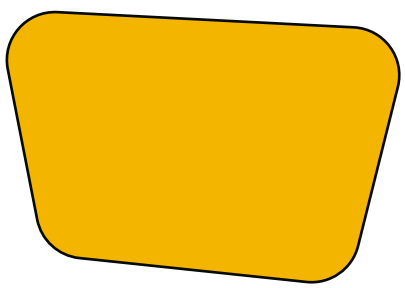
A frame's four corners are addressed by name - option and radius each, eight property paths. Five kinds, one mixed outline, and one polygon that shows where the addressing ends.



Rounded Inverse Bevel Inset Fancy



One rectangle, four kinds: rounded top-left at 28, bevel top-right at 28, inset bottom-left at 18, inverse bottom-right at 18. Per-corner addressing means the outline can disagree with itself, on purpose.



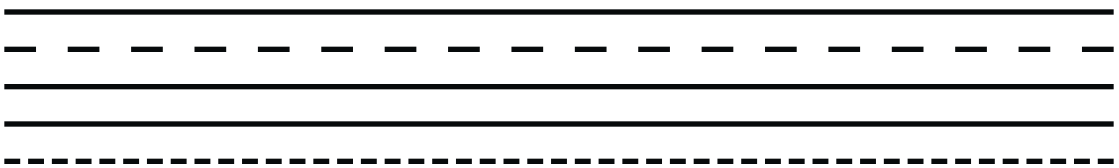
The same vocabulary on a polygon: the TopLeft slot rounds every corner uniformly - the four names are rectangle-shaped addressing, and IDML has no per-name mapping for a shape whose corners are not four. The BevelCorner written to TopRight is stored and round-trips; it steers nothing.

◪ *Polygon corners: only the TopLeft slot (or the global pair) drives geometry, uniformly across all corners; the other three names store and round-trip without rendering - B-23's recorded residual → Appendix A.*

Stroke anatomy



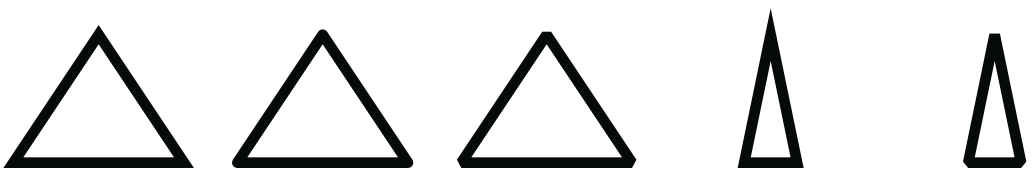
frameStrokeWeight, a ramp: 0.25 / 0.5 / 1 / 2 / 4 / 8 pt.



frameStrokeType: Solid, Dashed, Dotted, Dashed4-4 - then frameStrokeDashArray, a 6/3 pt override.



frameStrokeEndCap on dashed strokes: Butt, Round, Projecting - every dash segment wears the cap.

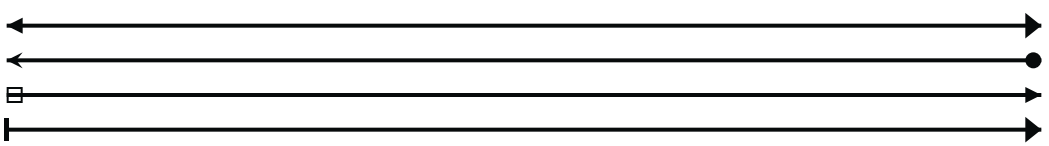


frameStrokeJoin on a 45-degree apex: miter, round, bevel. Then frameStrokeMiterLimit at work on a 24-degree spike: limit 12 keeps the point, limit 1.5 shears it to a bevel.



frameStrokeAlignment: inside, centre, outside - a 6 pt stroke within, astride, around the box.

frameStrokeGapColor + frameStrokeGapTint: the dash gaps filled with Lab Marigold at tint 100 (left) and 35 (right).



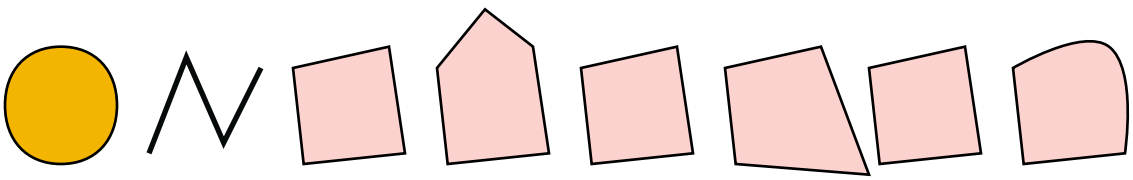
Arrowheads, start and end per line: Simple/TriangleWide, Barbed/CircleSolid, Square/Curved, Bar/SimpleWide - eight of the IDML ArrowHead kinds.

SPECIMEN NO. 68 · FRAMESTROKEWEIGHT/TYPE/DASHARRAY · ENDCAP · JOIN · MITERLIMIT · ALIGNMENT · GAPCOLOR/GAPTINT · START/ENDARROWHEAD

◪ *Kind boundaries on the wire: frameStrokeEndCap and frameStrokeAlignment are rectangle-addressed (caps shown on dash segments - a closed outline has no open end), and arrowheads are graphic-line-only → Appendix A.*

Topology and the pathfinder

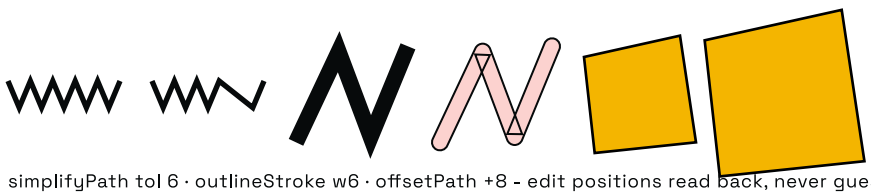
Each tile in the upper band holds the same shape twice - untouched at left, operated on at right. The lower band feeds overlapping pairs to every planar verb the wire declares.



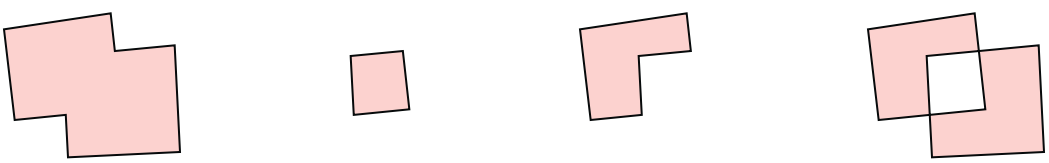
insertPath (closed Bezier blob + open polyline) · pathPointInsert · pathPointSet · pathPointCurveType



pathPointRemove · pathOpenAt · closePath · joinPaths (two open strokes welded into one)



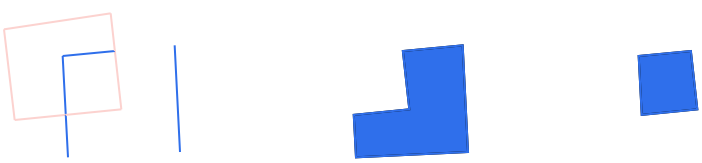
simplifyPath tol 6 · outlineStroke w6 · offsetPath +8 - edit positions read back, never guessed.



pathfinderBoolean: union · intersect · subtract · exclude (the bottom shape's paint survives)

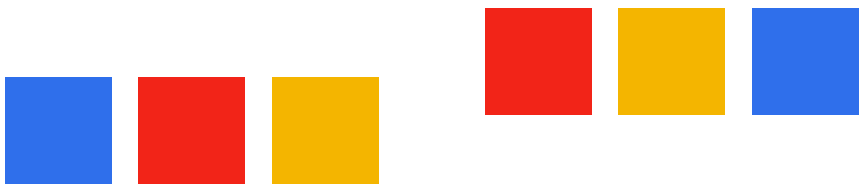


The region verbs, one fresh pair each: divide · trim · merge · crop (the top shape is the cookie cutter)

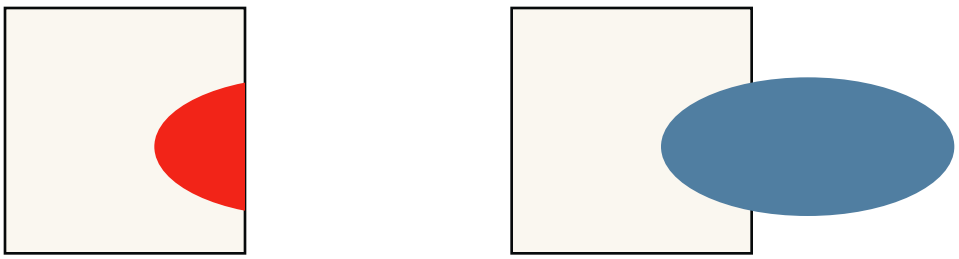


outline · minus back · faces, keeping only the overlap face 0-1#0 - read back, never guessed.

Groups, nests, and the dressed frame



The upper trio was grouped and moved 180 by 44 as ONE setGroupTransform; the lower trio was grouped and dissolved - members stay, the wrapper goes.



Left: an oval pasted INTO a frame - it is clipped at the container's edge, and deleteFrame refuses it until released. Right: the same nesting, then releaseFrom - the twin ends the page whole, overlapping its container unclipped.



The frame at left wears Exhibit Frame, an object style that did not exist when this chapter began. It was born Exhibit Panel, renamed live, and based on the fixture's Annotation Marker - which is where the tinted fill and hairline come from, because the definition door for object styles is closed today.

A second style, Scratch Object, ran the whole triple - created, renamed, deleted - and the checkpoint carries no trace of it: demonstrated, not resident.

SPECIMEN NO. 70 · CREATEGROUP / SETGROUPTRANSFORM / DISSOLVEGROUP · PASTEINTO / RELEASEFROM · OBJECTSTYLE CREATE/RENAME/APPLY + DELETE (TRANSIENT)

◪ *setStyleProperty answers UnsupportedProperty for the object collection (probed live) - an object style dresses frames only through basedOn; and B-18: a nested child leaves the scene tree and deleteFrame refuses it until released → Appendix A.*

9

Ink & Light

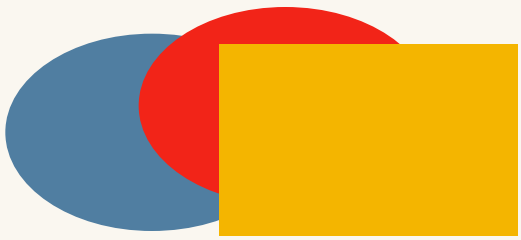
Ink is subtractive and patient; light is additive and quick. Where the two share a page, the page needs arithmetic — sixteen kinds of it.

The plate overleaf is one motif set sixteen times: two soft shapes and a marigold square, identical to the last point, except for the compositing arithmetic the square carries. The engine speaks the full InDesign blend vocabulary — Normal through Luminosity — and the plate names every member rather than sampling the flattering ones.

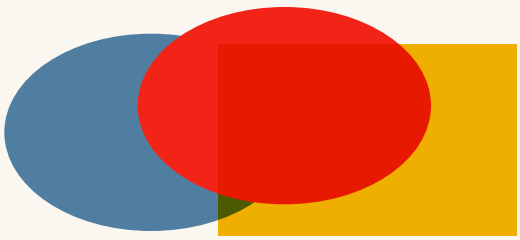
The closing page turns to light's other instruments. Two gradients are minted live on the wire and one is edited in place, to prove the ramp is data rather than ink. An opacity ramp steps a vermilion chip down into the paper. And two opacity masks — one reading luminance, one reading alpha — do what no IDML element can. The margin says so plainly, and the export ledger counts the loss out loud.

The sixteen modes

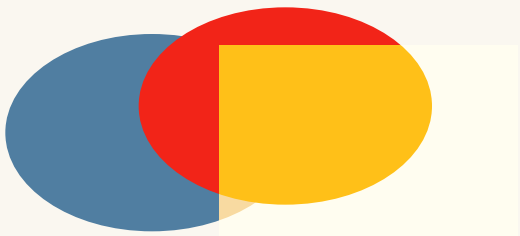
One motif, sixteen compositors. Two soft shapes and a marigold square, identical on every tile; only frameBlendMode changes.



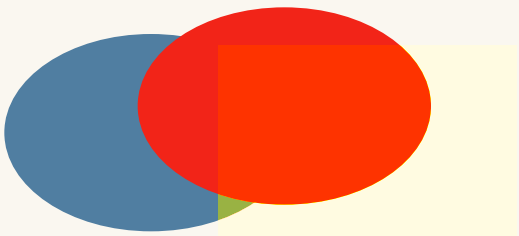
1 · Normal



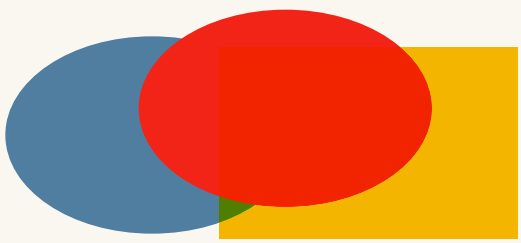
2 · Multiply



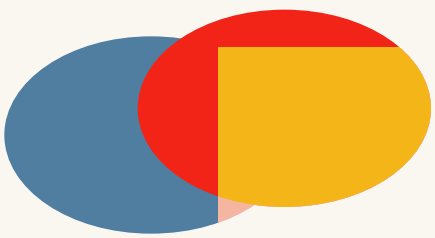
3 · Screen



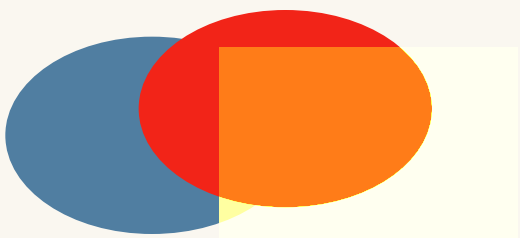
4 · Overlay



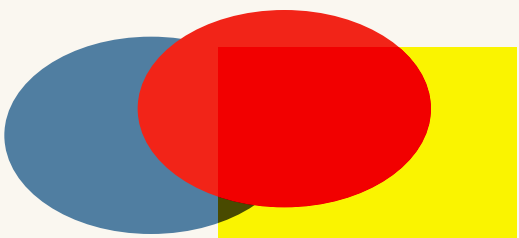
5 · Darken



6 · Lighten



7 · ColorDodge

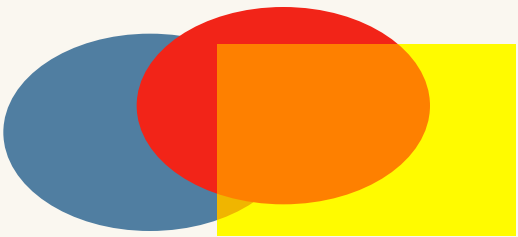


8 · ColorBurn

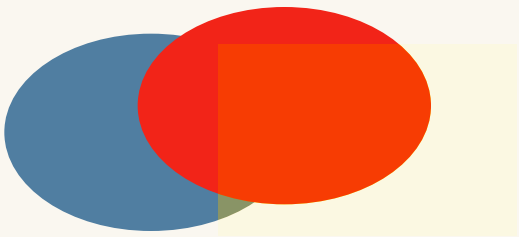
SPECIMEN NO. 81 · FRAMEBLENDMODE ×16 · NORMAL → LUMINOSITY · INSERTOVAL / INSERTFRAME / BATCH · D-PLATE

Hard Light to Luminosity

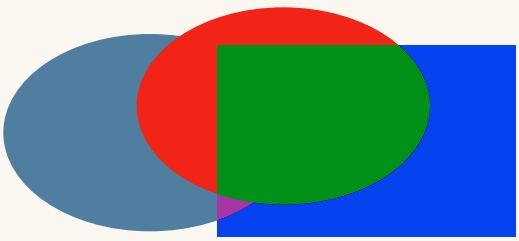
The second eight trade tone for colour: the last four blend hue, saturation, colour and luminosity rather than darkness.



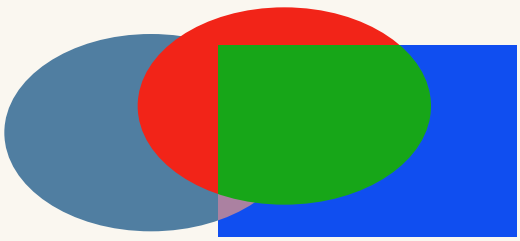
9 · HardLight



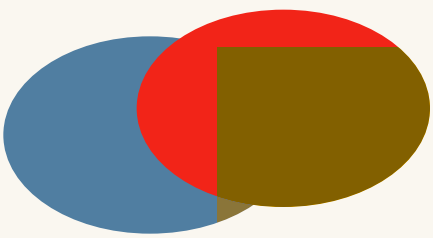
10 · SoftLight



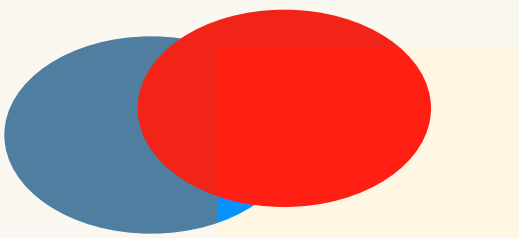
11 · Difference



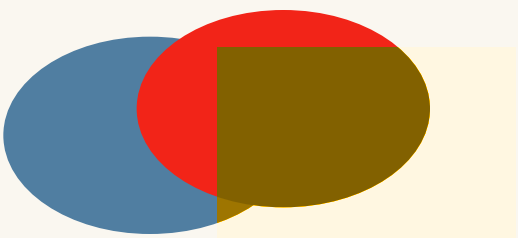
12 · Exclusion



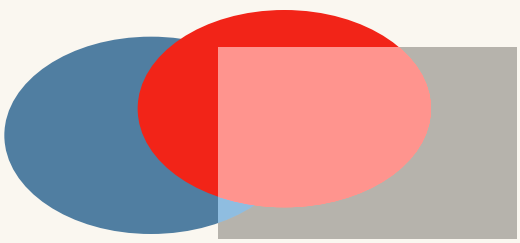
13 · Hue



14 · Saturation



15 · Color



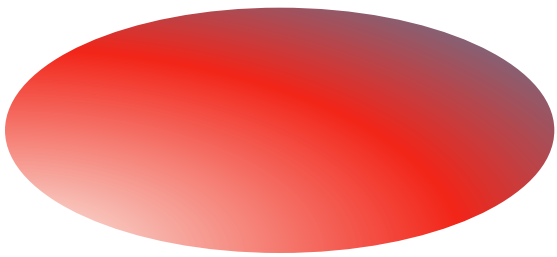
16 · Luminosity

Gradients, opacity, masks

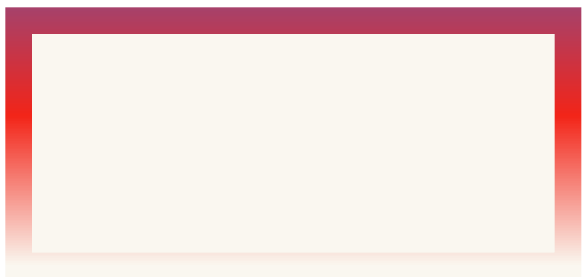
Everything on this page is light-work: two gradients minted on the wire while you watch, one of them edited after the fact; a ramp of vanishing vermilion; and two soft masks, one read by its luminance and one by its alpha.



Ink Dawn, linear — minted paper → vermilion → slate, then edited on press: the cold end you see is the editGradient, not the mint.



Ink Halo, radial fill.



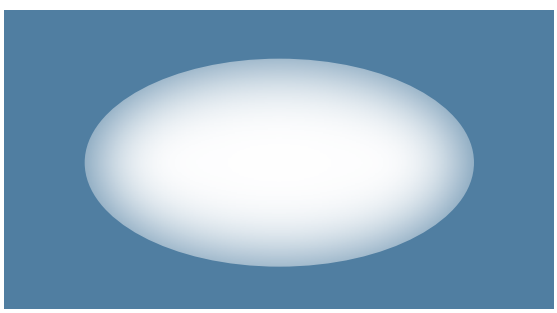
The dawn again, as a 10 pt gradient stroke.



frameOpacity · 100 · 80 · 60 · 40 · 20 · 10 — the same vermilion, surrendering to both grounds at once.



Luminosity mask — the Annual Ramp as artwork: vermilion survives the light end, dies in the dark.



Alpha mask, inverted — a feathered oval punches a soft window through the slate.

□ *The renderer owns a sweep/conic gradient paint, but no authoring door mints one: GradientSpec is Linear | Radial only, and conic colour reaches a page only through paged.web's bake. Recorded, not faked → Appendix A.*

□ *The masked pairs are certified on the CPU rasterizer, the fidelity path of record; GPU-lane soft-mask compositing arrived later, and canvases before it show these pairs unmasked → Appendix A.*

SPECIMEN NO. 82 · CREATEGRADIENT ×2 · EDITGRADIENT · DELETEGRADIENT · FRAMEGRADIENTFILL*/STROKE* · FRAMEOPACITY · APPLYOPACITYMASK ×2 · RELEASEOPACITYMASK · DEMONSTRATED, NOT RESIDENT

◪ *IDML has no opacity-mask element. The .paged container keeps both pairs verbatim; the IDML twin cannot, and the export ledger names every masked item as lost — loudly, by design → Appendix A.*

10

The Effects

Eight families of borrowed light and manufactured shadow, some sixty property paths between them — and one honest contact sheet.

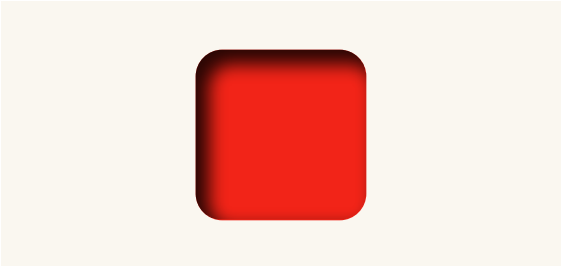
The verso overleaf is the contact sheet: the same vermilion square eight times, each carrying one family at full expressive settings — drop shadow, inner shadow, outer glow, inner glow, bevel and emboss, satin, feather, directional feather. The recto answers with two compositions that use them in anger, and gives the gradient feather, the ninth instrument, an exhibit of its own.

The closing page is the parameter table: for every family, a row of small squares that turn one knob at a time, so the reader can see what each path buys rather than take the panel's word for it. Two limits are recorded where they occur — satin stores an invert flag the rasterizer ignores, and a transparency group's raster bounds can clip a wide overhang that the PDF path keeps. Both stand in the margin and again in Appendix A.

Eight families, one square



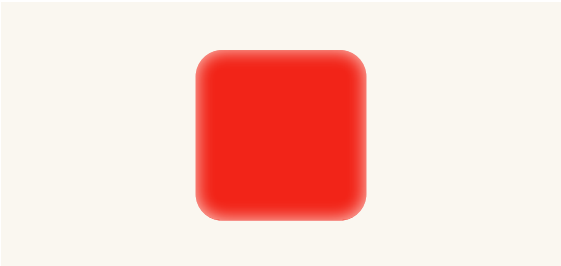
Drop shadow



Inner shadow



Outer glow



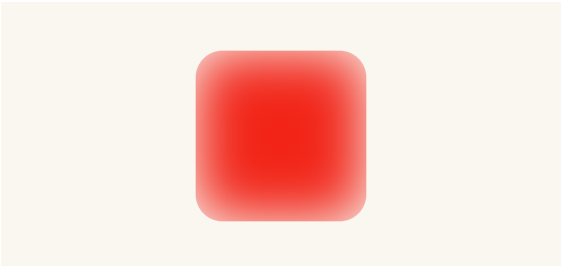
Inner glow



Bevel & emboss



Satin



Feather



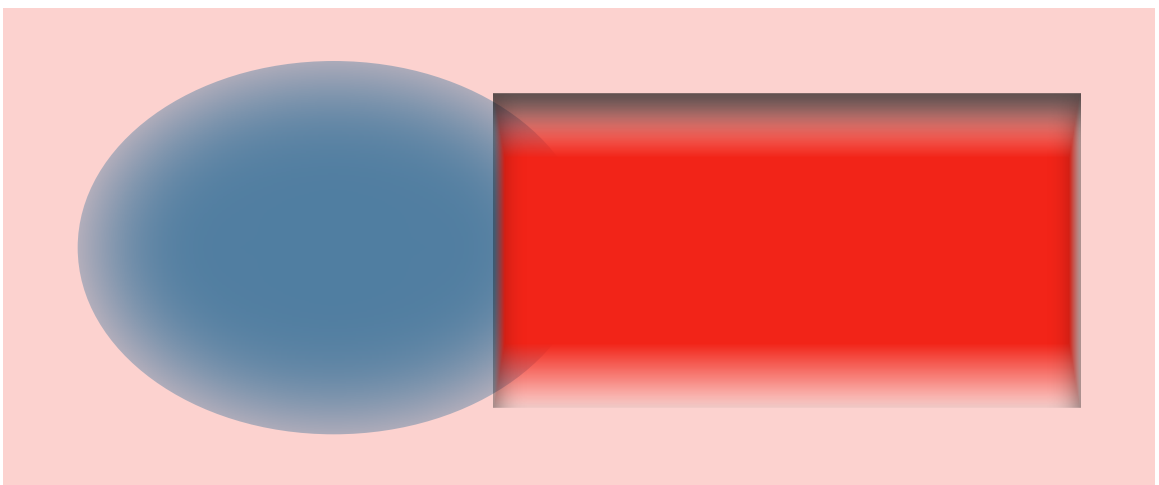
Directional feather

SPECIMEN NO. 85 · 8 FAMILIES, THE FULL BATTERY EACH · FRAME DROPSHADOW · INNERSHADOW · OUTERGLOW · INNERGLOW · BEVEL · SATIN · FEATHER · DIRECTIONALFEATHER

Effects in anger



Display type carrying drop shadow and a screened glow; the moon layers bevel and satin on one oval.

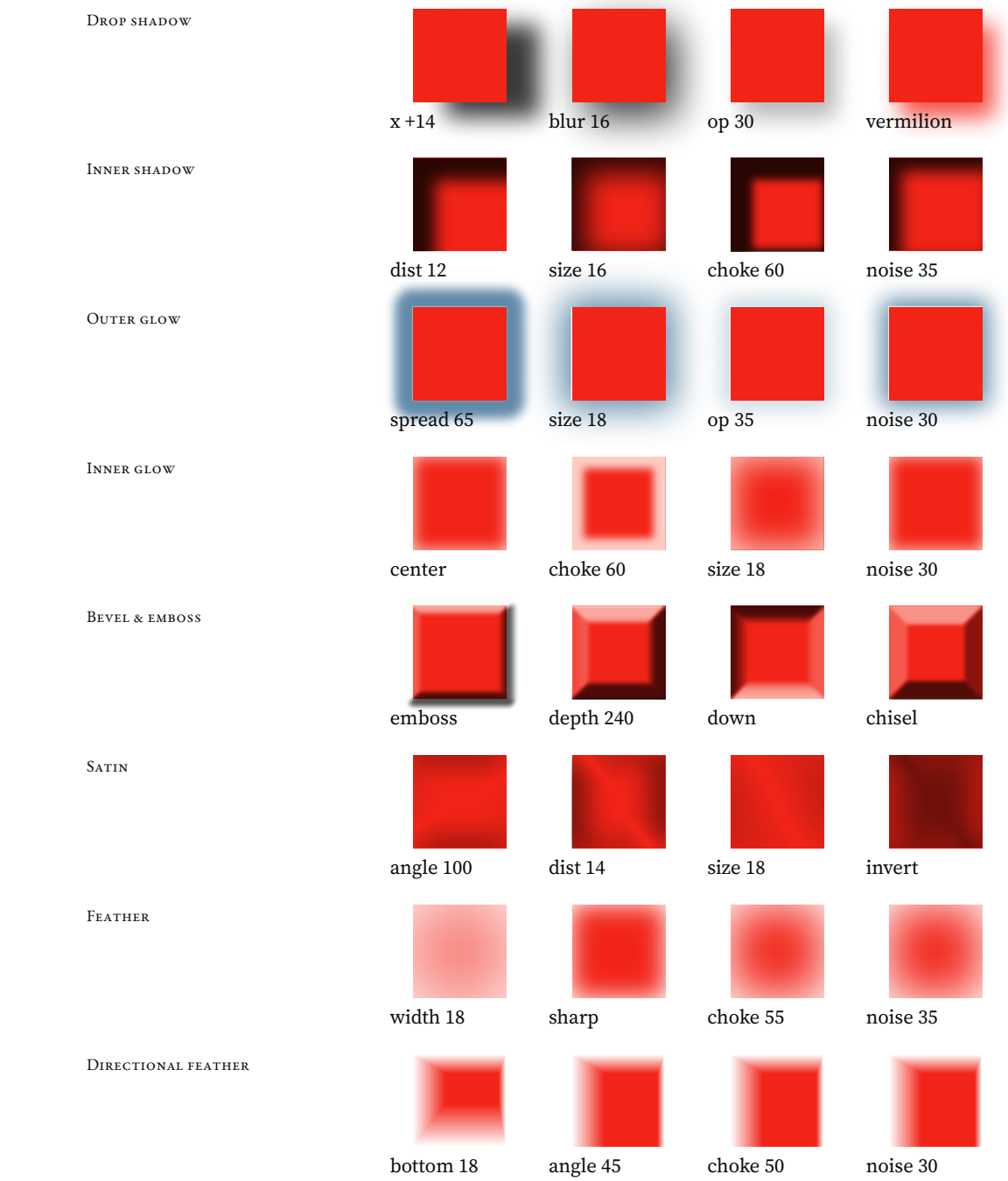


A feathered slate cloud and a vermilion veil losing its top and bottom edges to the directional feather, shaded from within.



frameGradientFeather — the fade is alpha, not ink: one bar, opaque to absent.

One parameter at a time



□ *A transparency group's raster bounds are the frame's geometry padded half a point: a wide stroke or a generous shadow can clip in raster output while the PDF path keeps it. A recorded defect, not a policy → Appendix A.*

SPECIMEN NO. 87 · 8 FAMILIES × 4 VARIANTS · FULL BATTERY + ONE KNOB EACH · EVERY FRAME-EFFECT PER-FIELD PATH, TWICE OVER

◪ *The satin row's registry entry still records invert as ignored by the rasterizer (W1.8); on this build the inverted square prints visibly darker — the flag reaches the compositor now, and it is the record that has drifted → Appendix A.*

11

The Colour

A document's palette is a contract with a press. This chapter reads the annual's seven inks off the live collection, mints and edits new ones, walks the library out as Adobe Swatch Exchange bytes and back, and ends at the prepress bench.

Annual Ink · process · CMYK 72/62/58/90 — the editorial black, built rich rather than flat K.

Paper Warm · process · CMYK 2/3/6/0 — the paper simulation every plate sits on.

Vermilion · spot · a real spot ink, CMYK build 0/85/90/5 — a named separation all the way to the plate; this composite previews its CMYK alternate.

Vermilion 20% · spot · a real CMYK-built tint, TintValue 20 on the vermilion build. Deliberately not an RGB tint: TintValue scales only through the CMYK resolve, so an RGB swatch carrying one would paint at full strength.

Slate · process · CMYK 65/45/30/10 — the annotation ink the spec labels wear.

Lab Marigold · process · Lab 78/15/82 — a true Lab primary, resolved analytically (D50 to D65 Bradford, then sRGB), no profile required.

Screen Blue · process · RGB 47/111/235 — the deliberate warning specimen: an RGB build in a print document, kept out of the brand group on purpose.

Minting ink

SEVEN INKS, AND NOW A PRESS GREY

Two swatches were created on this page and one of them was then edited in place. Both were born as the same flat 55% black; the right-hand chip's definition was replaced with a cooler 26/18/12/62 build through editSwatch, and every reference to it — including the words PRESS GREY above — repainted, because a chip holds an identity, not a colour. A third swatch, Scratch Cyan, and a Scratch Group ran the whole create-then-delete round transiently: demonstrated, not resident. The group holding the greys was born Annual Live and renamed Annual Live Inks through editColorGroup without disturbing its two members.



Press Grey (as minted) · CMYK 0/0/0/55 — the birth build, untouched.



Press Grey · CMYK 26/18/12/62 — the same identity after editSwatch.

OVERPRINT, DECLARED



Two identical Press Grey squares straddle the vermilion band. The left one knocks out, as every fill does by default; the right one carries frameOverprintFill and frameOverprintStroke, so at the press its grey lays down ON TOP of the vermilion instead of punching a hole in that plate.

SPECIMEN NO. 91 · CREATE/EDIT/DELETESWATCH · CREATE/EDIT/DELETECOLORGROUP (DELETES TRANSIENT) · CHARACTERFILLCOLOR · FRAMEOVERPRINTFILL/STROKE

◪ *Overprint is honoured plane-aware on the CPU composite; this RGB proof approximates the ink build-up, and the separations are the real evidence. Scratch swatch and group: demonstrated, not resident.* → *Appendix A*

There and back — the .ase loop

Three inks were minted for this page and gathered into a group called Annual Loop. The group then left the document: the export door serialised it to 204 bytes of Adobe Swatch Exchange, magic ASEF, version 1.0 — the same file a designer would hand to a colleague on another tool. Then all three swatches and the group were deleted; for a moment the palette genuinely did not hold them.

importSwatchLibrary read those bytes back. The chips below are painted from the REIMPORTED swatches, resolved by name — which is only possible because the originals are gone. Identity does not ride in an .ase: each ink returned under a fresh Color/u id. The names, the colour spaces, the channel values and the group all did, and Ledger Sea came back still a spot.

Ledger Red · process CMYK 10/95/90/2 · returned as Color/u11 (process)

Ledger Buff · process CMYK 6/12/35/0 · returned as Color/u12 (process)

Ledger Sea · born a SPOT, Lab 42/−18/−28 · returned as Color/u13 (spot)

The loop is the point: a palette is not locked in this file format. What leaves as an interchange library comes back as first-class document swatches, grouped as they left.

SPECIMEN NO. 92 · EXPORTSWATCHLIBRARY → ASEF · DELETESWATCH ×3 · DELETECOLORGROUP · IMPORTSWATCHLIBRARY · 204 BYTES ROUND-TRIPPED

The .ase format carries name, space, values and the spot flag — not tints and not alternates: Vermilion 20% would leave as a flat build, and Ledger Sea returns Lab-only, its CMYK alternate shed at the border. → *Appendix A*

The prepress bench

Everything on this page is a setting, not a shape. Each bench below applies one management op and then prints what the document reports back — the read is the exhibit. Where the effect lives at the proof or the press rather than in this composite, the label says so.

1 · SETCOLORSETTINGS

working CMYK profile — Paged Default CMYK (registered live, active: true)
rendering intent — Perceptual · black-point compensation — true
every CMYK chip in this chapter now resolves through this working space.

2 · SETPROOFSETUP

proof profile — Paged Default CMYK · simulate paper white — true
proof intent — RelativeColorimetric. Prepress state: the soft proof reads it; a composite page does not repaint.

3 · SETINKSETTING + SETUSESTANDARDLABFORSPOTS

Annual Ember (spot) — aliasTo Color/AnnualVermilion · convertToProcess false
two vermilions, one plate: the alias merges the separations at output time.
useStandardLabForSpots — true: Lab-primary spots now paint from the standard Lab book values.

4 · SETDOCUMENTDEFAULTS

default fill — Color/AnnualVermilion20 · default stroke — Color/AnnualSlate at 1 pt
the rectangle below was inserted with no property writes at all — it arrived wearing these defaults.





Annual Ember — the aliased spot, previewing through its CMYK build.

Born dressed: no fill or stroke was ever set on this rectangle.

SPECIMEN NO. 93 · REGISTERCOLORPROFILE · SETCOLORSETTINGS · SETPROOFSETUP · SETINKSETTING · SETUSESTANDARDLABFORSPOTS · SETDOCUMENTDEFAULTS · DOCUMENTMETA + INKS (LIVE READ-BACK)

Colour settings, proof setup, ink aliases and document defaults are app-level, non-undoable session state; their full effect belongs to the proof and the separations, and this composite carries only what repaints. → Appendix A

12

The Table

A table is not a grid of rectangles that happen to align; it is a structure the text owns, with a header that knows to say itself again. This chapter builds one taller than its frame and walks it across the gutter, then takes a cell apart, rebuilds the grid live, and dresses it in styles.

One table, two pages

The table below is taller than the frame that holds it, on purpose. Its story is threaded into a second frame on the facing page; the break falls mid-body, and the header and footer bands say themselves again on the other side of the gutter.

Table 2 — the annual's own page plan: every chapter of this document, the pages it owns, and how many leaves that is.

NO.	CHAPTER	PAGES	LEAVES
1	Front matter	1–10	10
2	Part I divider · Ch.1 Anatomy of This Book	11–18	8
3	Ch.2 The Letter	19–22	4
4	Ch.3 The Paragraph	23–26	4
5	Ch.4 The Style	27–32	6
6	Ch.5 The Story	33–40	8
7	Ch.6 Scripts of the World	41–44	4
8	Ch.7 The Apparatus	45–46	2
9	Ch.8 The Object	47–52	6
10	Ch.9 Ink & Light	53–56	4
11	Ch.10 The Effects	57–60	4
12	Ch.11 The Colour	61–64	4
13	Ch.12 The Table	65–70	6
14	Ch.13 The Picture	71–74	4
134 pages in 22 chapters — and this table itself occupies two of them.			

NO.	CHAPTER	PAGES	LEAVES
15	Part II divider · Ch.14 The Drawing Office	75–86	12
16	Ch.15 The Darkroom	87–94	8
17	Ch.16 The Ledger	95–102	8
18	Ch.17 The Long Read	103–108	6
19	Ch.18 The Catalog	109–114	6
20	Ch.19 The Manuscript	115–118	4
21	Part III divider · Ch.20 Imposition & Proof	119–126	8
22	Appendix — Limits, Index, Colophon	127–134	8
134 pages in 22 chapters — and this table itself occupies two of them.			

The cell, taken apart

Sixteen cells, one property each, every label naming the exact path that produced what you see: the fills, then two cells wearing appliedCellStyle — Annual TH and Annual TD Number — the four insets (read where each caption stands), the four vertical justifications, and the twelve per-edge stroke paths — colour, weight and tint for each of the four edges.

cellFillColor	cellFillTint 30	Annual TH	TD Number
cellInsetTop 16	cellInsetLeft 24	cellInsetBottom 16	cellInsetRight 24
TopAlign the default	CenterAlign centred	BottomAlign seated	JustifyAlign slack shared
top edge	bottom edge	left edge	right edge

Twenty property paths on one grid; the twenty-first, appliedTableStyle, dresses the long table overleaf. The EDGE tints paint — the pale right edge is the vermilion at 40 — and so does the FILL tint: the second cell is the same vermilion at 30.

SPECIMEN NO. 103 · CELLFILLCOLOR + CELLFILLTINT · CELLINSETTOP/LEFT/BOTTOM/RIGHT · CELLVERTICALJUSTIFICATION INCL. JUSTIFYALIGN · 12 × CELL EDGE STROKE PATHS · APPLIEDCELLSTYLE ×2

cellFillTint is stored per cell: a read answers 30, and the second cell paints the vermilion at that tint beside the full-strength first. Alternating row fills come from an applied table style (the ledger overleaf). → Appendix A

Rebuilding the grid, live

The operations log below is itself the table the operations ran on. It was minted four rows by three columns; a row and a fourth column arrived afterwards, live; a scratch row and a scratch column each existed for exactly two operations — created, then deleted — so the record shows them without the grid keeping them.

OP	WHAT HAPPENED	AT	COL
insertTable	minted this grid: four rows, three columns, poured full	—	
insertTableRow	this row arrived AFTER the pour — everything below it renumbered	at 2	new
deleteTableRow	ran on a scratch row that stood below this one for exactly two ops	transient	
deleteTableColumn	ran on a scratch fifth column — same two-op lifetime	transient	

the band bench	2 × 2, before
and after	still 2 × 2

The band bench: a header and a footer were inserted and removed again on this small table. Four real ops, net zero — the long table on the previous spread is where the bands stayed to repeat.

One structural door does not exist: a table inside a cell. The wire's insertTable takes a story address and carries no cell qualifier — checked against the generated wire types, not guessed — so a nested table cannot be authored live. Documents that arrive with one render it faithfully; this page simply cannot make one.

SPECIMEN NO. 104 · INSERT/DELETETABLEROW · INSERT/DELETETABLECOLUMN (DELETES TRANSIENT) · INSERT/REMOVEHEADERROW · INSERT/REMOVEFOOTERROW · SETCOLUMNWIDTH

Nested tables render from IDML; the wire has no door (insertTable addresses a story, never a cell). Scratch row, column and bands: demonstrated, not resident. → Appendix A

Dressing the grid

Two styles were minted for this page: a table style born Ledger and renamed Financial, and a cell style born TD Money, renamed TD Currency, based on the fixture's Annual TD Number. A scratch pair ran the whole create-and-delete round transiently. What a live-created style cannot yet do is carry arms — the definition door refuses the cell and table collections, so Financial is a name with a cascade position and nothing in its pockets. The switch below makes that visible instead of hiding it.

STYLE	REGIONS
Annual Table	header + body
cascade	resolved live

STYLE	REGIONS
Financial	header + body
cascade	resolved live

The same table twice. Left, appliedTableStyle names the fixture's Annual Table: header band, alternating fills, hairlines — all region defaults. Right, it names the live-created Financial: the dressing vanishes, because the style is real but empty.

Table 3 — the print order, closed out.

QTR	MILESTONE	RUNS	COST (EUR)
Q1	Specimen sections set	34	12 408.00
Q2	Corrections and reflows	18	6 512.50
Q3	Plates and proofs pulled	27	9 940.25
Q4	Bound copies delivered	55	21 780.75

The figures column is set in Table Number — right-aligned tabular lining, the fixture's own style — with appliedCellStyle naming Annual TD Number per cell. Decimal figures align on their tabular widths and the right edge.

SPECIMEN NO. 105 · CREATE/RENAME/DELETETABLESTYLE · CREATE/RENAME/DELETECELLSTYLE · APPLIEDTABLESTYLE ×3 · APPLIEDCELLSTYLE ×8 · SETSTYLEPROPERTY (CELL) REFUSED

◪ *setStyleProperty refuses the cell and table style collections (no refusal recorded…), and decimal tab stops are not addressable in cells, so the figures align right in tabular lining. → Appendix A*

13

The Picture

A photograph enters a document twice: as pixels to decode and as a claim about their source. Here one image runs six codecs, an EPS stays undecoded, and fitting is worked.

Alpine sunburst at sunrise · Sagui Andrea, Pexels · 2200 × 1469 px, 801,589 bytes of JPEG, inline



SPECIMEN NO. 110 · PLACEIMAGE + REPLACEIMAGEBYTES (INLINE) · APPLIEDOBJECTSTYLE: PLATE FRAME

One photograph, six codecs

Every exhibit on this spread is the same 800 × 534 photograph in an identically sized frame; only the encoding differs. All six went into the document as inline bytes and were decoded by the engine itself — a page this green is six codecs answering. The differences worth a loupe are qualitative and stated per label.



PNG · 682,620 bytes · 8-bit/channel RGB, lossless filters + DEFLATE. The reference print of the six.



WebP · 99,954 bytes · VP8 lossy, YUV 4:2:0 — an eighth of the PNG's bytes; the chroma carries the cost.



TIFF · 1,336,198 bytes · LZW, 8-bit RGB, little-endian — the prepress workhorse, lossless like the PNG.

Same pixels, five sizes: the byte counts on these labels are read from the committed files at build time, not typed. PNG, TIFF and BMP are lossless siblings — their renders should be indistinguishable. WebP trades chroma for an order of magnitude; GIF trades the whole palette.



GIF · 305,084 bytes · 87a, one adaptive 256-colour palette. Look at the apples' shoulders: continuous tone posterises where PNG keeps it smooth.



BMP · 1,281,654 bytes · Windows 3.x, 24-bit, uncompressed — every pixel spelled out; the largest file for the same picture.



CMYK JPEG · 520,203 bytes · baseline DCT, FOUR components, core's default CMYK profile embedded — decoded through the ICC lane, not assumed sRGB.

EPS · 987 bytes of EPSF-3.0 · recognised, NOT decoded — there is no PostScript interpreter, by declaration. What you see is the frame's own fill standing in: the honest render of content that exists but cannot be rasterised.

EPS is recognised, not decoded (no PostScript interpreter, by declaration): the frame renders its own fill rather than a false missing-image badge over content that is present. → Appendix A

The fitting model, and the lane that lasts

The same mountain ridge four times, in frames cut against its aspect on purpose. Each label names the properties that produced the geometry — the enum records the intent, the crops and the inner transform do the work.



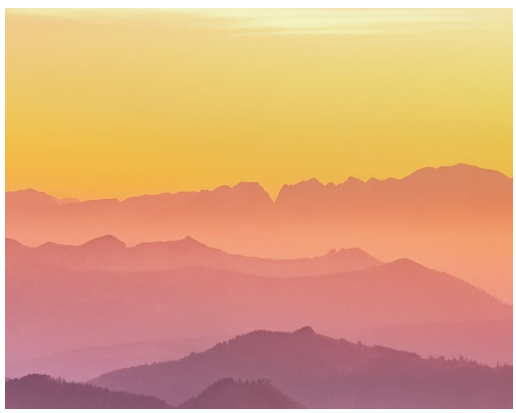
a · FillProportionally — cover. Crops 0 / -21.7 / 0 / -21.7: the overhang is negative crop, and the frame clips it, as InDesign does.



b · Proportionally — contain. Crops 13.9 / 0 / 13.9 / 0: the slack becomes bands of paper.



c · FitContentToFrame — the stretch, distortion and all. frameFittingReferencePoint: BottomRightPoint; frameAutoFit: true — both persisted intent for future fits and resizes.



d · imageContentTransform — a 2× cover, panned to the ridge. The inner transform maps image pixels into the frame; the frame's path clips the excess.





Left: born linked to nothing, it drew the grey placeholder, then was relinked. Right: the same start, then 99,954 bytes inline — the lane every photograph in this chapter rides.

SPECIMEN NO. 113 · FRAMEFITTINGTYPE ×3 · CROPS ×2 · REFERENCEPOINT · AUTOFIT · IMAGECONTENTTRANSFORM · REPLACEIMAGEBYTES INLINE + CLEAR

Fitting enums, the reference point and auto-fit are persisted intent — the geometry above comes from the crops and the inner transform; the cover fit's overhang is clipped by its frame, as InDesign clips it. *→ Appendix A*

Part II —

The Studios

Part I set type on a page the editor owns end to end. Part II walks the studios that rent space inside it: a drawing office, a darkroom, a ledger room — each a plug-in speaking to the same engine through the same wire, minting the same native objects a hand would. The next chapter belongs to paged.draw, and everything it leaves on these pages is anchors, swatches and strokes; nothing is a picture of a drawing.

14 The Drawing Office



The crest above arrived as a file of curves — annual-crest.svg, fed to the host importer registry and claimed by paged.draw. The importer lowered every contour through the same insertPath the pen tool commits, so what stands on this page is six native polygons wearing real swatches, not an embedded picture of a drawing.

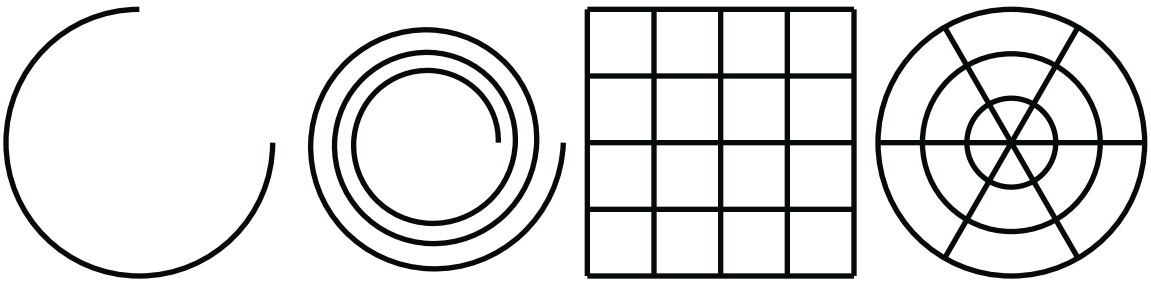
Two of those shapes are compound. The engine fills non-zero, and the importer delivers a compound path as sibling contours, so the annulus first painted as a solid coin; Compound Path > Make merged each pair into one polygon and re-wound the inner contour by nesting depth. The paper you can read through the ring is the whole argument of this chapter: a drawing tool that writes the document's own geometry, not a layer above it.

SPECIMEN NO. 115 · MEDIA.PAGED.DRAW.IMPORTER.SVG · INSERTPATH ×6 · MAKECOMPOUNDPATH ×2 · C-OPENER

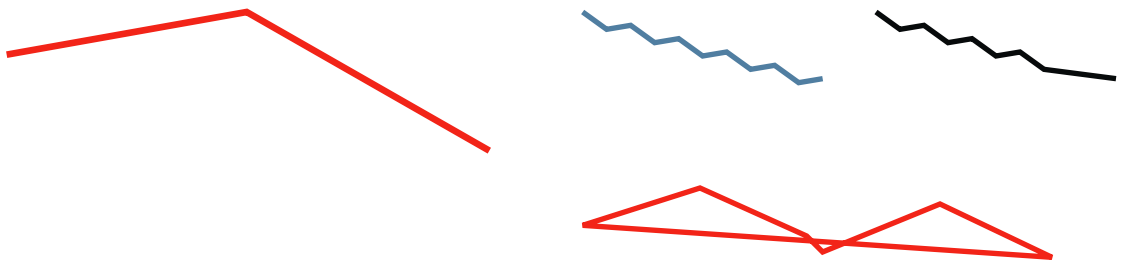
◪ *the importer's documented envelope is solid colours and geometry - gradients, <text> and <image> in an SVG do not lower; the crest was authored inside that envelope → Appendix A*

The drafting instruments

Four parametric generators, the pen, the anchor editors and the path surgery, all committing through the one insertPath lane. Every shape below is a native polygon the moment it exists.



Left to right: Insert arc, spiral, rectangular grid, polar grid — the parametric generators at their defaults, each re-seated into its tile. The arc and spiral carry real cubic handles; the grids are sibling lines.



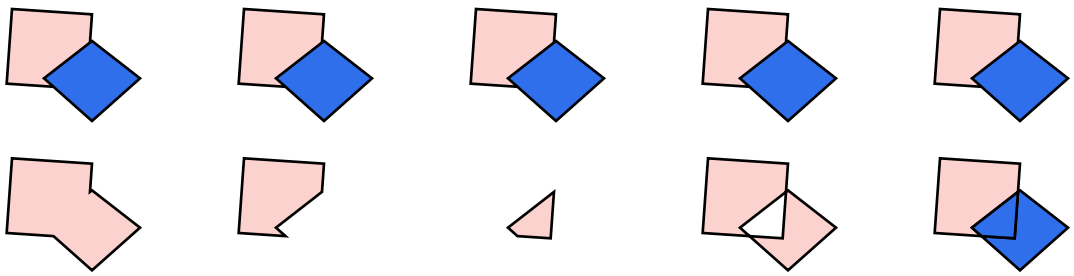
E

Band two, left: three pen clicks committed the open vermilion path - real pointer input through the live camera. The anchor editors then split, removed and smoothed a scratch path's anchors on the facing recto, every position read back from the engine between steps (the margin note says why not here). Right: the same jittery polyline twice, the second simplified at 3 pt tolerance; below, two open angles welded by Join and sealed by Close path.

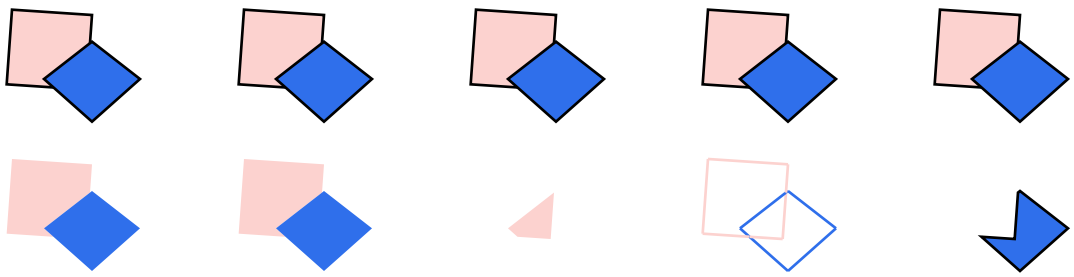
The anchor editors compare a page-local pointer with spread-stored anchors, one page width apart on this verso, so the surgery ran as a transient scratch on the recto; the insert commands take fixed defaults → Appendix A

The pathfinder wall

Ten planar verbs as paged.draw commands over a live selection. Each column holds the untouched scratch pair above and the same pair after its verb below - the operand order is the selection's, exactly as a menu invocation would read it.



Unite, Subtract, Intersect, Exclude - the shape modes; the first-selected quad keeps its identity and receives the result. Fifth column: Divide splits the pair into every planar piece.



Trim, Merge, Crop, Outline, Minus back - the region verbs, resolved over one planar arrangement with ownership following the z-order you see: the blue diamond sits in front, so it wins every contested face.

The engine's planar ops were shown raw in Chapter 8. This wall is the COMMAND lane over them: the same arrangement kernel, reached through the registry a menu bar, a palette and a shortcut all share - which is why every result here also answers undo as a single step.

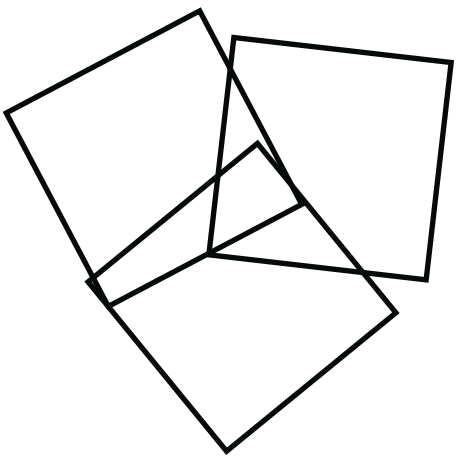
SPECIMEN NO. 117 · PATHFINDER ×10 (4 SHAPE MODES + 6 REGION VERBS) · 10 FRESH PAIRS, SELECTION-ORDERED

◪ *the arrangement runs in raw path space and composes at most 12 inputs / 256 faces; past either cap the engine refuses rather than truncates* → *Appendix A*

The builder and the bucket

Two ways to paint with an overlap. The Shape Builder unites what a drag sweeps; Live Paint treats the overlap's regions as fillable faces - derived, painted and released below with every face id taken from the engine's own planar answer.

A warm and a blue rectangle overlapped corner over corner; one Shape Builder drag from inside the first to inside the second swept both and committed a single union - the stepped silhouette above is ONE element now, wearing the first-swept warm fill, and one undo step.



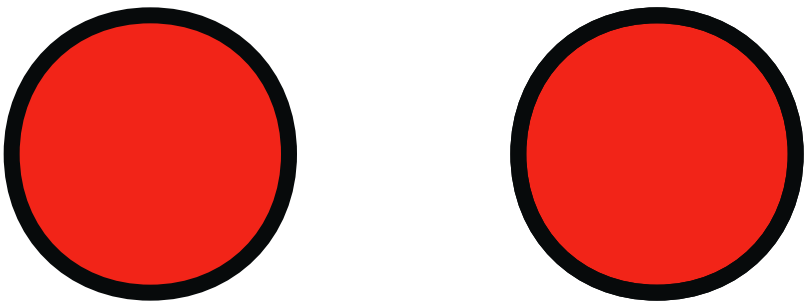
Three stroked, unfilled quads; their overlaps make 7 planar faces. A Live Paint group recorded the trio, 3 faces were filled by id and interior point from the engine's planar answer, Select faces put the 3 painted fills on the selection, and Release kept every stroke and fill while dropping the recipe. Nothing here holds a face object: each colour is a real polygon inserted over its region.

SPECIMEN NO. 118 · SHAPEBUILDER DRAG-UNITE · MAKELIVEPAINTGROUP · FILLLIVEPAINTFACE ×3 · SELECTLIVEPAINTFACES · RELEASELIVEPAINT

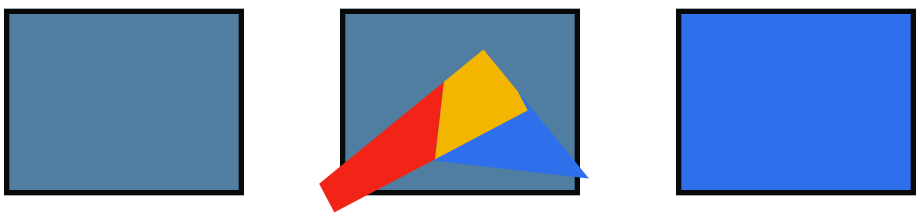
◪ *Live Paint here is regenerable, not live: no persistent faces, gap tolerance or edge strokes, and a painted face lands above the strokes that bound it. On this verso the fills landed a page width off and were re-homed → Appendix A*

The appearance stack

An IDML frame owns one fill and one stroke. The stack below is plugin metadata plus a bake: the left blob is LIVE (stack on the envelope, front-most layer on the frame), the right one is the same stack lowered to a carrier group of single-paint items.



Two identical blobs, each carrying +1 fill and +2 strokes over their own paint, one stroke reordered by Appearance: Reorder layer. The left is live metadata; the right was group-baked into a carrier plus 3 derived single-paint layers. They render alike - that is the bake's whole claim, and release restores the carrier exactly.



Three tiles, one saved style. Apply linked all three; a direct recolour of the first marked it overridden without breaking its link; Redefine pushed first blue, then slate, through every link it found. The third tile broke its link after the blue pass - it keeps that appearance while its neighbours moved on. A style that only copied values would be a clipboard; the link is the feature.

SPECIMEN NO. 119 · APPEARANCEADDFILL/ADDSTROKE/MOVELAYER · BAKEAPPEARANCE (B-24 CARRIER GROUP) · SAVEGRAPHICSTYLE/APPLY/REDEFINE/BREAKLINK

◪ *the front-most-layer bake is NOT a composite of N fills - it lowers the top opaque layer to the frame's single fill/stroke slots; the carrier-group bake is the faithful form, at 2 undo steps because insertPath mints ids a batch cannot address in itself → Appendix A*

Paint, banked

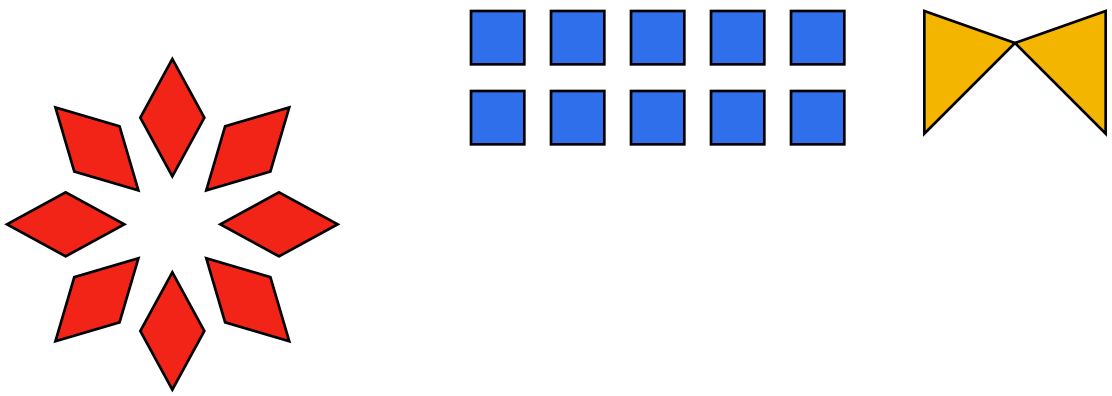
This page was designed to carry paged.draw's paint lane: the two gradient commands minting their stop swatches, a brick pattern field of real copies, and the dash presets over a selected line. In isolation, every one of those exhibits runs and verifies. Against this document — ninety pages of authored content — the engine hangs indefinitely while composing this page once the exhibits are in place, and it kept hanging as the exhibits were trimmed: without the dotted presets, with a quarter of the pattern, with lines instead of paths.

A specimen book that hid that fact would be a brochure. The paint lane's coverage is claimed nowhere on this page; its commands are proven by the isolated run this module cites, and the hang is the finding the margin carries.

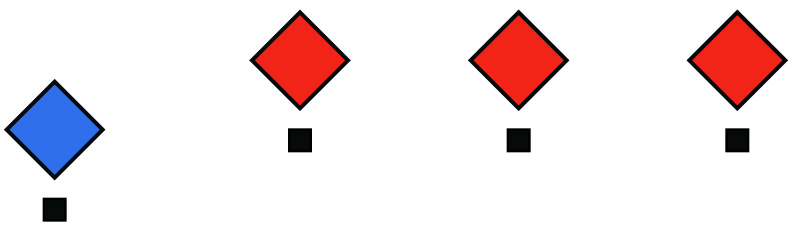
◪ *Composing this page with the paint exhibits in place hangs the engine indefinitely against the grown document (deterministic; survives every content trim; green in isolation). The full module is 07-paint.ts, runnable solo. → Appendix A*

Repeats and symbols

Two ways to say 'again'. A repeat transforms copies of a source and stays re-plannable; a symbol is a named definition whose placed instances follow a redefine - until one breaks its link on purpose.



Left: one petal, radially repeated 8 times on a 40 pt ring, every instance rotated to follow - then EXPANDED, so the rosette is now plain artwork with no recipe. Middle: a chip gridded 3x3, then re-planned to 5x2 by Update; every instance is a new element each time, which is what re-plannable honestly costs. Right: a pennant and its mirror twin over a vertical axis. Each make was ONE undo step - one batch inserts, binds, paints, links and groups.



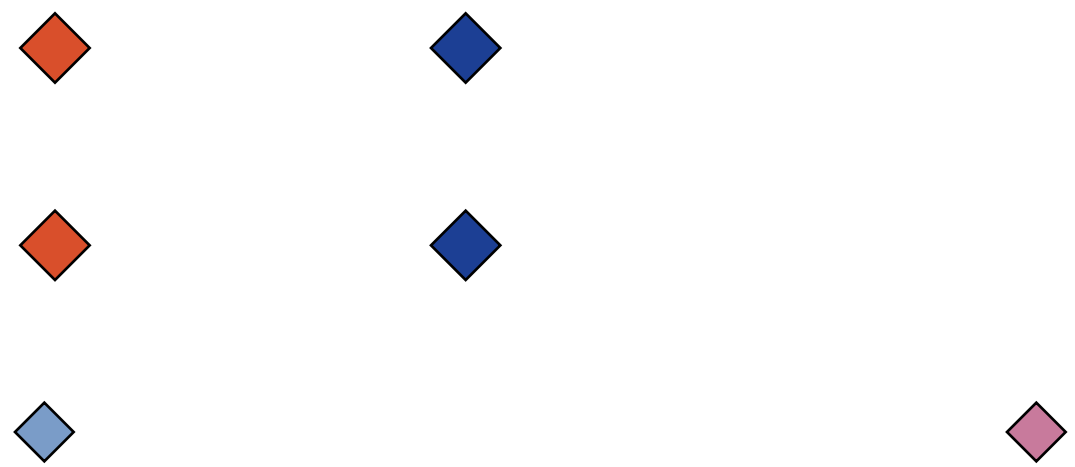
One badge defined from the two source shapes at far left and placed three times - each instance re-emitted native geometry with a per-leaf link. The source then turned blue, and no badge followed: an instance is STATIC (core has no symbol node and no duplicate op), so a new look travels only through Redefine's rebuild - which this engine refuses at the instance-group re-create, the same one-batch dissolve-and-regroup seam the margin note records for Update. The first instance also broke its link: it now would not follow even a successful redefine.

SPECIMEN NO. 121 · MAKERADIAL/GRID/MIRRORREPEAT · UPDATE · EXPAND · DEFINESYMBOL · PLACE ×3 · BREAKSYMBOLLINK

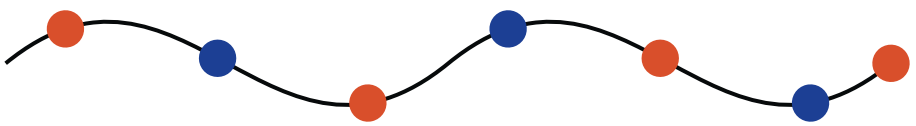
symbols refuse text and have no symbol sets, nine-slice or 3D mapping; Redefine rides the same refused batch and is not run here → Appendix A

Blends and objects on a path

Interpolation and distribution. A blend mints NEW intermediate shapes between two keys; objects on a path moves the shapes you already have and mints nothing - two arranging verbs with opposite manners.



Top: Specified Steps placed exactly 5 intermediates, then Reverse spine read the spine from the far end - the warm shape now finishes the run. Middle: Specified Distance fit 8 intermediates at one per 18 pt of spine, then Expand kept them all and dropped the recipe. Bottom: Smooth Color derived 78 steps from the keys' own colour distance and shingled them into the full-width ribbon. A typed count over 200 is refused; a derived one clamps - a typo is not data.



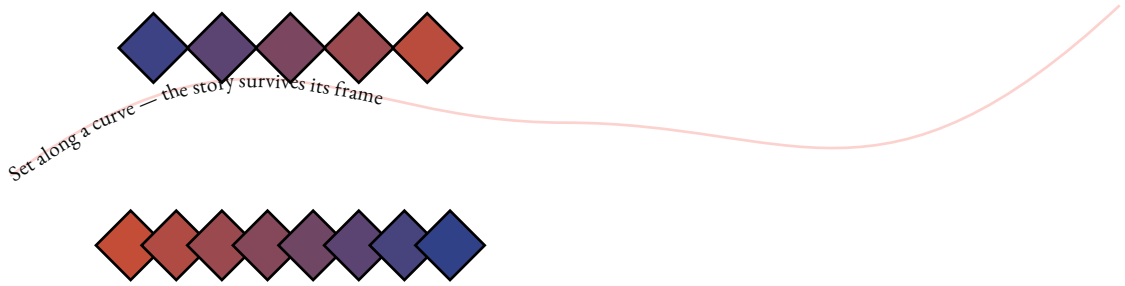
Seven dots drawn in a row below, then distributed along the wave - the dots ON the path ARE the originals, moved by one transform write each (ids intact, nothing minted, one undo step). Update slid the whole run 26 pt from the path's start, re-placing every dot from its recorded HOME transform rather than from where it stood - which is what makes it idempotent, and what makes Release an exact restore rather than an inverse - proven on a scratch trio that went home and left the page (demonstrated, not resident).

SPECIMEN NO. 122 · BLENDSELECTED STEPS/DISTANCE/SMOOTHCOLOR · REVERSEBLENDSPINE · EXPANDBLEND · MAKEOBJECTSONPATH · UPDATE · RELEASE (TRANSIENT)

A blend refuses keys whose subpath structure differs, and text frames outright; its intermediates minted one page width off on this verso and were re-homed by one transform batch → Appendix A

Type on a path, corners and kin

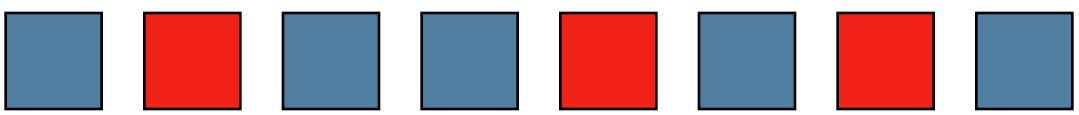
A story freed of its frame and flowed along a curve; rectangle corners restyled live; and a selection that finds its own family by fill.



The sentence above flows an ordinary story along an ordinary path - attachTextToPath took the story this page typed, styled and then freed by deleting its frame (no op mints a bare story; that workflow IS the door). A scratch pair also[illegible]e exhibit.



Rounded, Bevel, Fancy - three of the five live-corner presets, each an eight-write batch onto the rectangle's real frameCornerOption and frameCornerRadius properties, 12 pt radius all round.



Eight tiles, three of them vermilion. One was selected by hand; Edit > Select same > Fill grew the selection to 97 elements — every carrier of that swatch in the document, not just these tiles. Selection mutates nothing; this count is the evidence.

SPECIMEN NO. 123 · ATTACHTEXTTOPATH (CENTERPATHTYPE) · DETACH (TRANSIENT) · CORNERSROUNDED/BEVEL/FANCY · SELECTSAMEFILL

◪ *corner presets are shown on RECTANGLES - the engine accepts the corner properties on more kinds but renders them where an enclosed corner exists; and only RainbowPathEffect renders for path type, so the attach offers no effect knob → Appendix A*

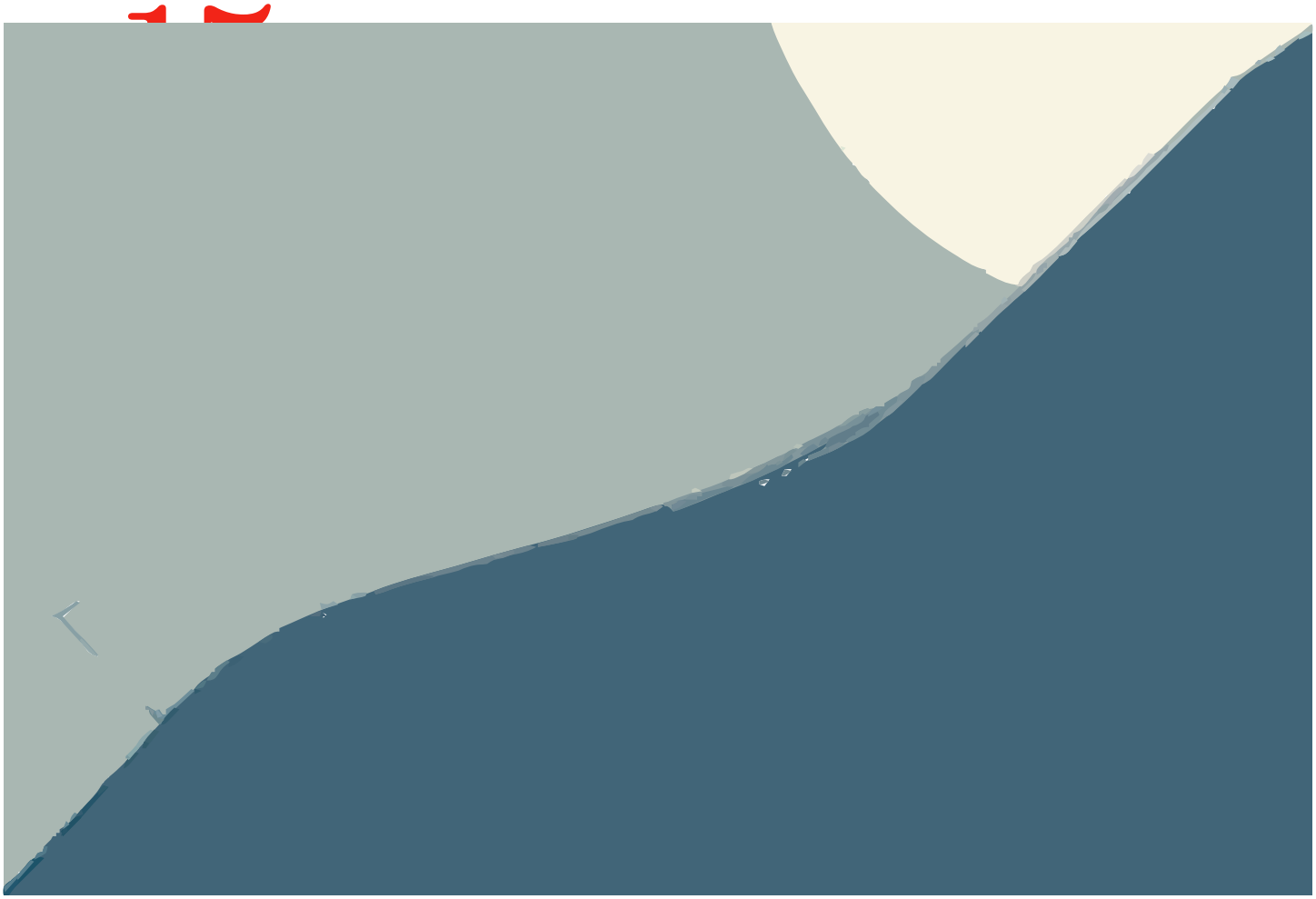
PLATE: THE MACHINE TRACE

The band above is not a photograph. pexels-1103970 (a study in sand curves) went into the document as inline bytes, and paged.draw's imageTrace ran in-editor through the wasm tracer - colour clustering, boundary walking, spline fitting - minting 94 filled native polygons whose holes are re-wound for the engine's non-zero fill. The source image was then deleted; what prints, exports and re-edits is vector.

Chapter holdings, read from the container itself (listPagedParts): under paged/media.paged.draw/ this document now carries blend.json, graphic-styles.json, live-paint.json, objects-on-path.json, repeat.json, symbols.json - 6 recipe part(s), the drawing office's ledger of styles, symbols, fields, rings, blends, dots and painted faces.

SPECIMEN NO. 124 · IMAGETRACE (WASM, IN-EDITOR) · MAKEOPACITYMASK/RELEASE (TRANSIENT) · LIST-PAGEDPARTS INVENTORY

◪ *The trace is one-shot and fills-only (demonstrated, not claimed); its regions minted one page width off on this verso and were re-homed in one batch. Releasing an opacity mask restacks it on top* → *Appendix A*



Place writes a link and a fitting; ingest decodes the placed bytes into the plugin's session; adjust runs the registered kernel chain on the GPU; composite paints the result over the frame as a Stage-A scene layer. Four of the five stations touch only the session. Commit is the exception and the point: the exported pixels re-enter the document through replaceImageBytes, the inline lane the container round trip preserves.

The kernel contact sheet

One photograph, ten tiles, ten kernel families. Every tile was ingested through the raster importer, resampled to 480 × 320 through the panel's lanczos3 resize, given exactly one family's representative kernel, exported through the bundle's own encoder and committed back as inline bytes. The apples are Tom Swinnen's (Pexels); the differences are the registry's.



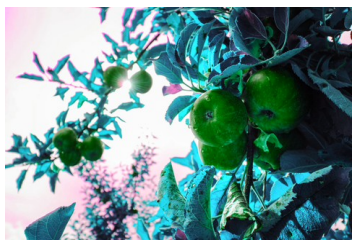
EXPOSURE · ADJUST.EXPOSURE
+1.6 EV
69,201 B COMMITTED



BRIGHTNESS+CONTRAST · ADJUST.BRIGHTNESS_CONTRAST
+0.25 ×1.9
72,628 B COMMITTED



SATURATION · ADJUST.SATURATION
×2.6
70,800 B COMMITTED



HUE ROTATION · ADJUST.HUE_ROTATE
+120°
62,920 B COMMITTED



BLACK & WHITE · ADJUST.BLACK_WHITE
SIX-WEIGHT MIX
49,812 B COMMITTED



POSTERIZE · ADJUST.POSTERIZE
4 LEVELS
615,733 B COMMITTED

THE ROSTER — EVERY REGISTERED KERNEL, 131 IDS IN 14 FAMILIES, READ FROM REGISTRY/KERNELS.YAML AT BUILD TIME

```
adjust (21)  exposure brightness_contrast levels saturation hue_rotate invert_rgb
white_balance vibrance hue_saturation dodge_burn color_balance black_white
posterize threshold photo_filter lut1d lut3d gradient_map channel_mixer levels_rgb
selective_color
conv (12)  box gaussian_h gaussian_v unsharp emboss find_edges motion lens bilateral
shape smart_sharpen radial
compose (27)  normal normal_gamma multiply screen overlay darken lighten
color_dodge color_burn hard_light soft_light difference exclusion hue saturation
color luminosity linear_burn linear_dodge darker_color lighter_color vivid_light
linear_light pin_light hard_mix subtract divide
math (16)  linear invert add sub mul div add_const mul_const abs sign neg min max
clamp min_const max_const
```


◪ *A kernel run is session state. Each tile here persists only because its adjusted pixels were exported and committed back as inline bytes; reopening the file re-reads those bytes, not the kernels.* → *Appendix A*

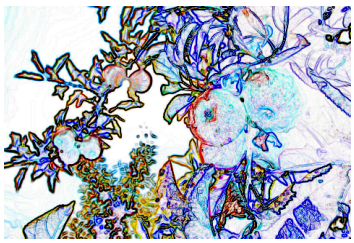
Each caption names the family, the dispatch id from the kernel registry, the panel setting, and the byte count of the committed export — the number that proves the tile you are looking at is document data, not a live preview. Kernels run once, in the session; what persists is what came back through the loop.



GAUSSIAN BLUR · CONV.GAUSSIAN_H + CONV.GAUSSIAN_V
Σ 4 PX
24,135 B COMMITTED



UNSHARP MASK · CONV.UNSHARP
AMOUNT 2.5
96,831 B COMMITTED



EDGE DETECTION · CONV.FIND_EDGES
STRENGTH 1
615,733 B COMMITTED



FILTER GALLERY · GALLERY.HALFTONE
6 PX DOTS AT 45 DEG
615,733 B COMMITTED

Eight tiles are parameter kernels — the slider writes a value, the export runs the chain, nothing touches the session source until the loop commits. The edge-detection and halftone tiles are effect kernels: their buttons apply destructively into the engine-held source, journaled in the plugin's own undo. Both kinds leave the document untouched until replaceImageBytes — which is why the loop, not the kernel, is this chapter's law.

```
band (4)  copy extract set_alpha broadcast_alpha
bool (4)  and or xor not
cast (2)  premultiply unpremultiply
rel (6)  eq ne lt le gt ge
geom (10)  flip_h flip_v rotate90_cw rotate90_ccw crop warp_backward rotate_bilinear
move_selection offset mosaic
gen (9)  solid checker linear_gradient radial_gradient pattern angular_gradient
reflected_gradient diamond_gradient noise
gallery (12)  kuwahara posterize_edges glowing_edges halftone grain diffuse
crosshatch bas_relief threshold_ink stained_glass glass texturizer
morph (2)  dilate erode
rank (3)  median3 despeckle dust_scratches
resample (3)  nearest mitchell lanczos3
```

SELECTIONS — THE MASK, MADE VISIBLE



THE MASK ABI

A selection is an engine-side coverage field over the ingested pixels. At dispatch it is lowered to a per-tile r16float window and bound at the kernel ABI's group 2 — every pointwise kernel composites mix(backdrop, result, mask), so one mask gates the whole adjustment vocabulary without any kernel knowing.

Add, subtract and intersect are max, min and multiply on the same field; feather is a gaussian pass over the mask, not the image.

What ran on this build: marquee: 0.0% at 0,0 0×0; quick selection grew to 0.0%. The quick selection mask (0.0% of the image, feathered) gated a saturation push to 2.8, and the masked result was exported and committed — 357,027 bytes of inline JPEG. Where the photograph above is loud, the mask said yes; where it is quiet, the same dispatch was arithmetic multiplied by zero.

The figures in this caption are the engine's own Selection readouts (bounds and coverage), read from the panel on this run — measured, not typed.

◪ *The selection itself is session state — a coverage field over ingested pixels, with no .paged representation. What persists is the masked result, committed as inline bytes. → Appendix A*

RETOUCH — PAINTING FROM THE IMAGE

The clone stamp and the healing brush are the brush with a different paint layer: the clone copies a window of the image from an alt-clicked anchor; the heal runs the same stroke and then tone-matches it in the gradient domain, so the patch takes on the destination's light. Content-aware fill synthesises a marqueed region from the rest of the picture, coarse to fine, every written pixel copied — never averaged.



BEFORE · AS SHOT · 168,209 B INLINE · TWO RETRIEVER PUPPIES, CHEVANON PHOTOGRAPHY, PEXELS



AFTER · CLONE + HEAL MOVE FLOWERED TURF INTO THE UPPER MEADOW; CONTENT-AWARE FILL LIFTS A CLUSTER · 240,887 B COMMITTED

The anchor click is a separate interaction from the stroke — alt-click sets the source and deposits nothing, because painting the anchor is the one thing a retoucher never wants. The heal's correction is the membrane field matching the source–destination mismatch on the dab's boundary; where the window runs off the image it falls back to a plain clone and says so in the panel. Attribution on this page is measured twice: each landing is read from the panel's status line, and the committed export is byte-compared against an identity baseline taken before the first stroke.

SPECIMEN NO. 134 · CLONE STAMP · HEALING BRUSH · CONTENT-AWARE FILL · ALT-CLICK ANCHOR, STROKE, MARQUEE, COMMIT

◪ *Strokes live in the plugin's session and its own undo journal, not in the document's history; the after frame persists only as the committed inline bytes.* → *Appendix A*

PAINT AND THE RASTER WORD

Painting needed no new kernel: a dab is a solid composited through the requested blend with stroke coverage as the mask, and erasing is one alpha write. Type is the fifth feature to fall out of the same mask ABI — a shaped glyph run IS a coverage field, so a word composites exactly like a brushstroke that happens to spell something.



THE STROKES AND THE WORD DID NOT LAND ON THIS LANE; THE TILE CARRIES THE PHOTOGRAPH UNPAINTED, AND THIS CAPTION IS THE RECORD OF THAT.

HOW A STROKE LANDS

Arm the tool, hover, drag: spacing carries leftover arc length across pointer samples, so a fast drag still paints a stroke rather than a row of dots. Parameters freeze at pointer-down. Strokes commit per gesture into the active layer, journaled in the plugin's bounded undo.

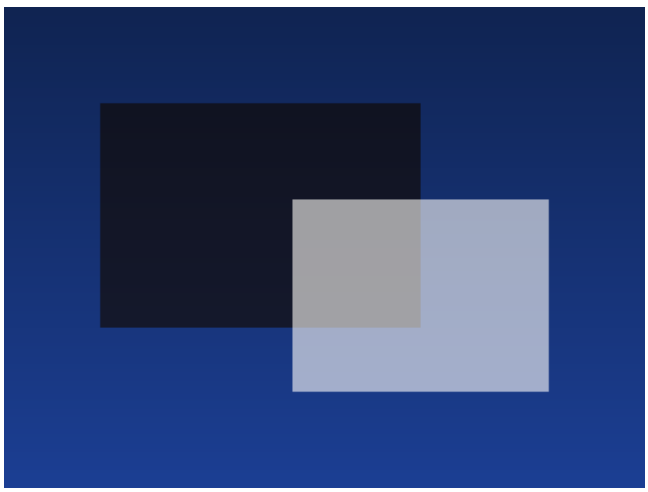
The word is shaped by the face's own tables through harfrust — joining scripts come out joined — and the face comes from the document's registry, never the network. A glyph the face lacks is counted and left undrawn.

SPECIMEN NO. 135 · BRUSH · PENCIL · RASTER TYPE (HARFRUST-SHAPED) · STROKES → EXPORT → REPLACEIMAGEBYTES

◪ *Paint and raster type are PARTIAL rows, demonstrated but unclaimed: strokes commit per gesture (Stage-B per-drag preview is deferred), and type is one run on one line — no wrapping, no text object.* → *Appendix A*

PSD — THE FILE THAT MUST SURVIVE

A PSD is somebody's working file, so the rule is preservation first: the reader retains every block it parses, unknown blocks opaquely, and a zero-edit save is byte-identical — an invariant proven in the plugin's own gates over a synthesized corpus and a 64-case property test, PSB included. What follows is the annual's own three-layer fixture through that machinery.



ANNUAL-LAYERS.PSD · 27,654 B ON DISK · COMPOSITE COMMITTED AS 481,099 B OF INLINE PNG

THE PARSED RECORDS

Backdrop — opacity 100%
Plate — opacity 78%
Signal — opacity 63%

Edited through the panel: Signal: 63% → 30%. The edited PSD exported as annual-layers.psd, 27,654 bytes — a record edit carried on the retained parse.

The canvas shows the import-time flatten; a layer edit lands in the EXPORTED file, not in a live re-composite — the panel states this limit where the edit is made.

Three layers — Backdrop (normal), Plate (multiply), Signal (screen) — assembled by the crate's own spec builders, mask-free by the layer-import tier's documented envelope. The preservation claim on this page is a citation, not a re-run: byte-identity at zero edits is the gate image-psd holds in its own test suite, and what this page adds is the editor reach — list, edit, export, through the panel a designer actually touches.

SPECIMEN NO. 136 · ANNUAL-LAYERS.PSD VIA THE RASTER IMPORTER · LAYER LIST · OPACITY EDIT · PSD EXPORT

◪ *PSD structural edits are record edits on the retained parse: visible in the exported file, not on the canvas (the import-time flatten stands). Round-trip proof spans the synthesized corpus; a real-Adobe corpus is the registry's named confidence step.* → *Appendix A*

THE DARKROOM LOOP, CLOSED

1 · Place. The frame takes a link and a fitting; the photograph goes in as inline bytes (263,842 B as shot).

2 · Ingest. The raster importer decodes the file into the paged.image session, bound to the selected frame.

3 · Adjust. Panel parameters drive the registered kernel chain on the GPU — here a 480 × 320 resample, +0.9 EV, saturation 1.5. The document is untouched.

4 · Export. The bundle's JPEG encoder re-encodes the adjusted pixels and the host delivers the file as a download — 71,973 bytes on this run.

5 · Commit. replaceImageBytes writes the exported bytes back into the frame — inline image data, the lane every checkpoint and the IDML round trip preserve.



STEP 1 — PLACED, AS SHOT



STEPS 2–5 — CORRECTED, COMMITTED

WHAT THIS CHAPTER DOES NOT CLAIM

Apply is a preview. The panel's Apply composites a Stage-A scene layer over the frame — the fastest way to see a correction, and session state by design. Reload the document and the layer is gone; that is why every exhibit here committed its pixels before its module returned.

The brush-library lane exists; its fixture cannot. The .abr reader is real — built clean-room against a measured corpus of 3,215 presets, a 24-variant warning channel for what it declines — but no redistributable .abr ships with this book: the published corpus references are licence-blocked for committed test assets, and authoring a fake library to press the button would demonstrate nothing true. The loader stays unpressed here.

Tiles are level-0. The plugin's tile provider serves the renderer full-resolution level-0 tiles; the buffer graph's mip pyramid stays engine-side, not yet across the wasm boundary. This chapter's exhibits never depended on it — the committed bytes are the document's own, decoded by the engine like any placed image.

Three stated limits: Apply's composite is session-only; the .abr lane ships no redistributable fixture (none authored, none pressed); the tile provider is level-0 only. → *Appendix A*

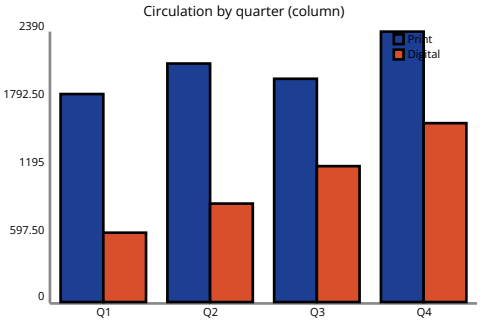
16

The Ledger

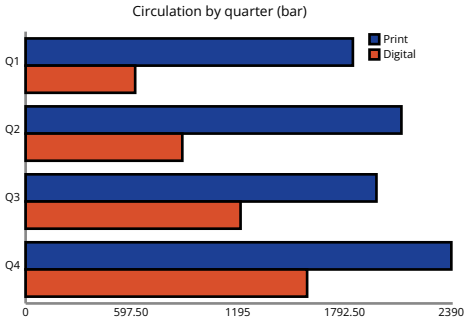
A spreadsheet in a page document leads two lives, and this chapter keeps them apart. The live grid that recalculates under your fingers is session state; what persists is the workbook and everything the plugin lowers to native content: real tables, charts drawn as real paths. The pages that follow lower ten charts, pour a range, spill a formula, and walk a day-book across a gutter.

The chart wall

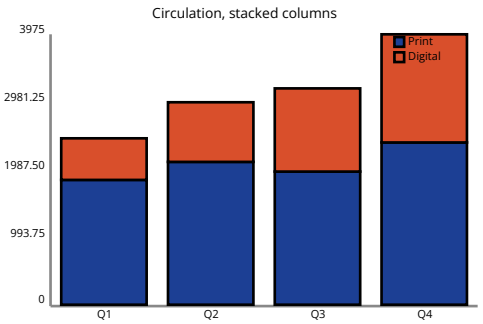
One workbook, ten charts — the engine's whole kind set, lowered one by one. Every bar, wedge, ring and spoke on this spread is a native path, and every axis number is a text frame; there is no picture of a chart anywhere in this document.



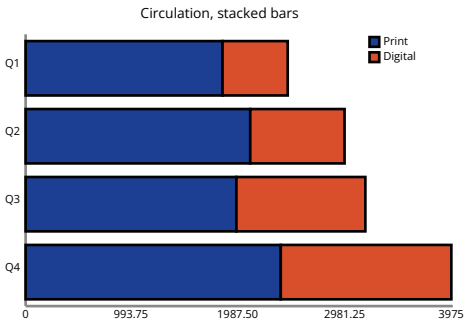
column — 12 native paths, 12 label frames



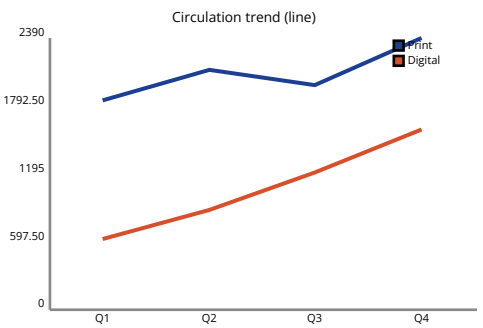
bar — 12 native paths, 12 label frames



stackedColumn — 12 native paths, 12 label frames

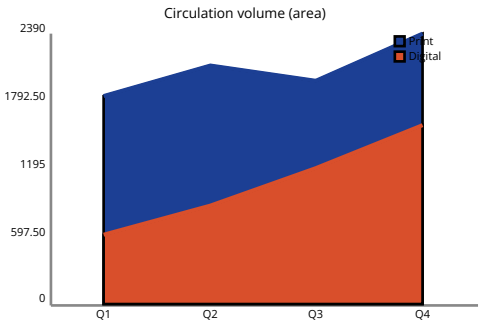


stackedBar — 12 native paths, 12 label frames

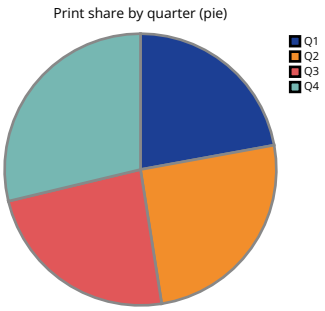


line — 6 native paths, 12 label frames

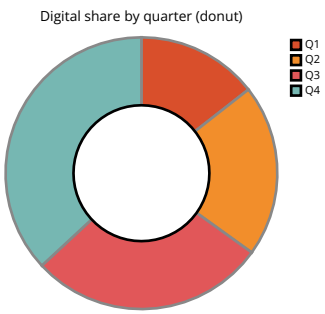
SPECIMEN NO. 151 · ANNUAL-CHARTS.XLSX VIA THE K-5 HOST FILE PICKER · LOWERCHARTTOFRAME (COMMAND LANE, CHART #0) · COLUMN · BAR · STACKED-COLUMN · STACKED-BAR · LINE



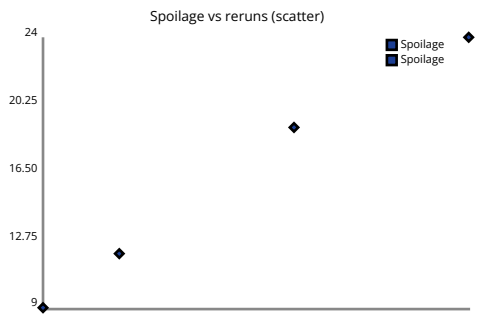
area — 8 native paths, 12 label frames



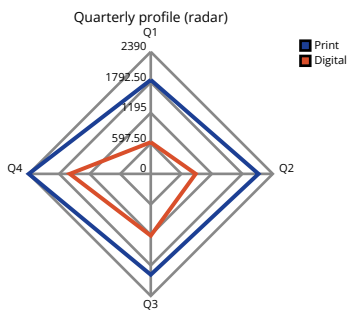
pie — 8 native paths, 5 label frames



donut — 9 native paths, 5 label frames



scatter — 8 native paths, 8 label frames



radar — 12 native paths, 12 label frames

How to read the wall: each specimen is the plugin's own lowering, scaled uniformly by one transform per chart — the geometry is the engine's, the slot is the page's. The colours are document swatches the lowering minted at content-addressed ids, so two charts sharing an axis grey share one swatch.

SPECIMEN NO. 152 · PER-CHART PANEL LOWERING ×5 · AREA · PIE · DONUT · SCATTER · RADAR · PLOTTERS → CHARTGEOMETRY IR → INSERTPATH + INSERTTEXTFRAME

The native pour

The quarterly circulation block — Data!A1:E5 of the chart wall's workbook — lowered to a native table. The engine computes the range in Rust; the bundle inserts a frame, then a real <Table> whose column widths were measured through this document's own font shaper, then pours every cell. It exports to IDML as a table because it IS a table.

Quarter	Print	Digital	Spoilage	Reruns
Q1	1840	620	3.2	12
Q2	2110	880	4.1	19
Q3	1975	1210	2.8	9
Q4	2390	1585	5	24

How the bundle finds what it made: a just-inserted EMPTY frame's story is invisible to the hit-test door — the text hit path needs content — so the lower resolves its story by diffing the stories collection across the insert. Only after the pour does hitTest answer at the frame's centre, and that is the door the chart label pours and the chain lowering use. Two read doors, split by emptiness; this page states the split rather than papering over it.

SPECIMEN NO. 153 · LOWERTOFRAME — DATA!A1:E5 · INSERTTEXTFRAME + INSERTTABLE + CELL POUR · COLUMNS VIA HOST.TEXT.MEASURESTRING · SHOW/HIDEGRIDINFRAME

◪ *The page lowering reads the workbook without its styles (getRangeLowered), so its cell fills and text colours do not reach this table. The styled read (getRangeStyled) is a separate door, deliberately not used here.* *→ Appendix A*

Formulas, spill, and the roster

3	Total	
2	Product	
5	TotalProduct	

The lowered range prints RESULTS: the SUM as 5 and 10, the concatenation as SumProduct, the uppercase call, and the spill range 10–40 that four cells inherited from one anchor. The formatted value is the value — the grid and the page share one format engine, so what you audited live is what went to press.

Table — the function library, by family, counted from the registry at build time.

FAMILY	FNS	REPRESENTATIVE CALLS
agg	9	AVERAGE · MIN · MAX
array	16	SEQUENCE · TRANSPOSE · UNIQUE
database	12	DSUM · DCOUNT · DCOUNTA
date	25	NOW · TODAY · DATE
fin	16	NPV · IRR · XNPV
info	17	ISBLANK · ISNUMBER · ISTEXT
lambda	8	LET · LAMBDA · MAP
logical	11	IF · AND · OR
lookup	16	VLOOKUP · HLOOKUP · INDEX
math	50	MMULT · SUM · PRODUCT
stat	47	MEDIAN · MODE · STDEV.S
t2misc	6	AGGREGATE · SUBTOTAL · ROMAN
text	30	LEN · LEFT · RIGHT
263 functions across 13 families — no row, no dispatch.		

SPECIMEN NO. 154 · IMPORTXLSX (COMMAND, HOST PICKER) · =SUM · =UPPER · =SEQUENCE(4,1,10,10) SPILL · LOWER A1:C6 — RESULTS ON THE PAGE · REGISTRY/FUNCTIONS/*.YAML ROSTER

Importing this workbook REPLACED the chart wall's workbook as the container part: workbook persistence is a per-plugin singleton (the LAST imported workbook travels), not per-frame. The chart wall survives regardless — its charts are native art. → *Appendix A*

A day-book across the gutter

1	1840	620
2	1847	633
3	1854	646
4	1861	659
5	1868	672
6	1875	685
7	1882	698
8	1889	711
9	1896	724
10	1903	737
11	1910	750
12	1917	763
13	1924	776
14	1931	789
15	1938	802
16	1945	815
17	1952	828

Twenty-four delivery days, three columns, all of it typed as three SEQUENCE anchors and spilled by the engine. The frame on this page is deliberately too short: the table breaks mid-body and walks across the gutter.

The header band was added AFTER the lowering, through the same insertHeaderRow every native table answers — the strongest evidence the plugin's output is not plugin-private.

SPECIMEN NO. 155 · =SEQUENCE ×3 → A8:C31 · LOWER → RESIZEFRAME → LINKFRAMES · INSERTHEADERROW ON THE LOWERED TABLE

◪ *The sheets engine has a pagination lane of its own (lowerPaginatedToChain) that no editor command reaches yet, so this spread shows the host chain threading the lowered native table, and claims only that. → Appendix A*

18	1959	841
19	1966	854
20	1973	867
21	1980	880
22	1987	893
23	1994	906
24	2001	919

The continuation — the same story, walking on across the gutter. (No header band could be added on this run; see the chapter notes.) The chain holds the whole day-book — nothing is overset past the last frame.

Round trips

Everything below happened while this page was being built, and the receipts carry the measured values of this very run: the sort and its principled refusal, the replacement that recalculated its dependents, the captured cell style, the exported workbook, and the part the container actually holds.

```
sortRange A1:A2 desc — Sorted (values lane)
sortRange C3:C6 (spill) — Sorted.
findReplace "Sum" → "Total" — inputs rewritten; B1&B2 recalculated to TotalProduct
copy/paste — degraded honestly: [media.paged.sheet] pasteAtSelection: the clipboard
is empty
styleFromCell — document cell style minted: Cell style
exporter — sheet-02-formulas.xlsx: 3053 bytes, magic "PK\u0003\u0004" (a valid OOXML
zip)
listPagedParts paged/media.paged.sheet/ — 2 part(s):
paged/media.paged.sheet/workbook.name, paged/media.paged.sheet/workbook.xlsx
```

The refusal above is a feature: a range whose cells are spill output cannot be reordered without corrupting the references that produced it, so the engine declines with its reasons instead of shuffling values it would immediately recompute. And the part listing is the chapter's opening promise kept — the workbook travels with the document, and the container will say so to anyone who asks it.

SPECIMEN NO. 157 · SORTRANGE (+ THE SPILL REFUSAL) · FINDREPLACE · COPYSELECTION · PASTESELECTION · STYLEFROMCELL · EXPORT CENTER → .XLSX DOWNLOAD · LISTPAGEDPARTS — WORKBOOK.XLSX

◪ *The sheetFromDataset command opens the datasets panel; no governed dataset was published in this session, so the consumer flow shows its honest empty state and sheet.data.consumer is not claimed. → Appendix A*

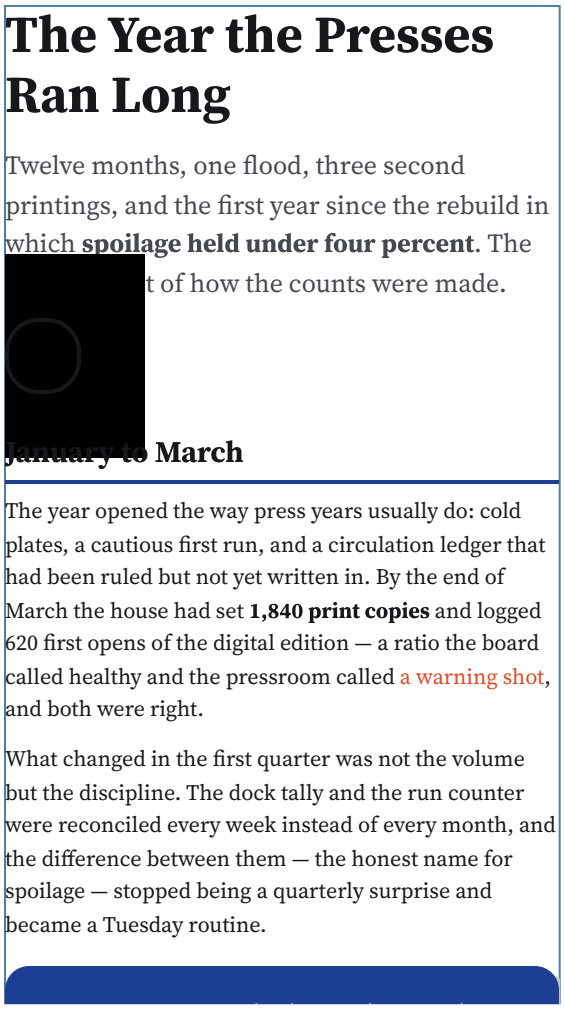
17

The Long Read

HTML and CSS are a writing system for documents, and this chapter treats them as content: an article laid out by a real browser engine and printed beside its own source. The pages that follow thread one source across many frames, break a forty-row table across three, and close on the asymmetry: a live render is session state, and only what is baked to native content survives.

Source beside render

The frame on the left is an ordinary rectangle whose document metadata carries this year's long read as HTML and CSS. The column on the right prints an excerpt of exactly what the envelope holds. The engine that laid the article out is a real browser stack, compiled to wasm, drawing into the frame.




```
// x-paged:media.paged.web — the envelope (v1)
{ "v": 1, "data": {
"html": /* 3687 bytes, excerpt: */
<article id="annual-longread">
<h1>The Year the Presses Ran Long</h1>
<p class="standfirst">Twelve months, one flood, three s…
the first year since the rebuild in which <strong>spoil…
percent</strong>. The long account of how the counts we…
<div class="badge"></div>
<h2>January to March</h2>
"css": /* 1216 bytes, excerpt: */
body { font-family: "Source Serif 4", serif; font-size:…
h1 { font-size: 26px; line-height: 1.15; margin: 0 0 10…
h2 { font-size: 15px; margin: 18px 0 6px 0; border-bott…
p { margin: 0 0 8px 0; }
em { font-style: italic; }
"options": { "media": "print", "overflow": "clip" } } }
```

Rendered live: the engine submitted a scene layer that core composes inside the frame. That paint is session state — the envelope beside it is what the document keeps.

SPECIMEN NO. 171 · INSERTWEBFRAME (ONE BATCH) · SETPLUGINMETADATA MEDIA.PAGED.WEB · RENDER-WEBFRAME → C-1 SCENELAYER · 5087 ENVELOPE BYTES

The live render is a C-1 scene layer and will not survive this chapter's checkpoint reload; the envelope in document metadata will. The chapter's last spread makes that asymmetry the whole point. → Appendix A

One source, two flows, four frames

The article declares flow-into: article-flow; its route notes declare flow-into: sidebar-flow. Frames are the real Regions consumers here — the chain below is persisted in the source's envelope in the document, and this page reads that record back as its proof.

The Year the Presses Ran Long

Twelve months, one flood, three second printings, and the first year since the rebuild in which **spoilage held under four percent**. The long account [redacted]ounts were made.

source — flow-into: article-flow

Route notes

The Alpine routes carried 3,410 print copies, the year's heaviest load, and lost none to weather. Harbour carried 2,905 and lost one bundle to the September flood.

The Plateau — six years on the print rolls — now opens more digital editions than

flow-into: sidebar-flow — the named flow, routed by threadWebFlowNamed

[redacted]

January to March

The year opened the way press years usually do: cold plates, a cautious first run, and a circulation ledger that had been ruled but not yet written in. By the end of March the house had set **1,840 print copies** and logged 620 first opens of the digital edition — a ratio the board called healthy and the pressroom called a warning shot, and both were right.

What changed in the first quarter was not the volume but the discipline. The dock tally and the run counter were reconciled every week instead of every month, and the difference between them — the honest name for spoilage — stopped being a quarterly surprise and became a Tuesday routine.

> A copy is counted when it leaves the dock. Everything else this house believes follows from that sentence.

April to June

flow-from consumers 1 and 2 — the primary chain, in selection order

SPECIMEN NO. 172 · THREADWEBFLOW ×2 · THREADWEBFLOWNAMED · UNTHREADWEBFLOW · RENDERWEBFLOW — EVERY FLOW GROUP · CHAIN VERIFIED VIA THE X-PAGED:MEDIA.PAGED.WEB LABEL

The flow renders above are scene layers — session state that the chapter checkpoint will drop. The CHAIN is not: it rides the source envelope in the document, and reopening the file re-renders it on any engine-capable lane. → Appendix A

The fragmentation ladder

Two sources, two rungs: a forty-row fares table split between body rows with its two-row header repeating, and a card column split between a card's own blocks — except the image, which moves whole.

Carriage fares, by route and zone

Route	Zone	Bundle (25)	Single copy	Carrier
name	ring	per bundle	per copy	contracted
Alpine North	1	4.10	0.22	Bergpost
Alpine North	2	4.60	0.24	Bergpost
Alpine North	3	5.30	0.27	Bergpost

Route	Zone	Bundle (25)	Single copy	Carrier
name	ring	per bundle	per copy	contracted
Alpine North	4	6.10	0.31	Bergpost
Alpine South	1	4.10	0.22	Bergpost
Alpine South	2	4.70	0.25	Bergpost
Alpine South	3	5.45	0.28	Bergpost

Route	Zone	Bundle (25)	Single copy	Carrier
name	ring	per bundle	per copy	contracted
Alpine South	4	6.30	0.32	Bergpost
Harbour East	1	3.80	0.20	Quayline
Harbour East	2	4.20	0.22	Quayline
Harbour East	3	4.85	0.25	Quayline

Carriage fares, one source, three frames (reading order: top left, top right, lower left). The split falls BETWEEN body rows only, and both header rows — the blue route line and the orange units line — say themselves again at the top of each continuation. Column widths may re-resolve per frame: the documented v0 caveat, content-loss-free; rows past the last frame are reported overset, never dropped silently.

Alpine North

The heaviest route on the rolls and the oldest: four zones, two carriers, and a driver who has never once conceded a bundle to snow. Carried **1,760 copies** this year without a single late delivery.

The zone-four leg was resurveyed in June and shortened by nine kilometres, which the fare card will notice next year even if the mountains do not.

Alpine South

The quieter twin: **1,650 copies**, one re-route around the April roadworks, and the network's best bundles-per-stop figure for the third year running.

Its zone-three fare rose five cents in autumn; nobody wrote in, which the office chooses to read as understanding.

Route cards, one source, two frames: a straddling card splits between its own blocks, recursively — the pressmark image is atomic and moves whole rather than tearing.

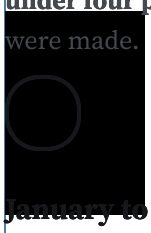
SPECIMEN NO. 173 · FARES-TABLE.HTML — 2-ROW THEAD, 40 BODY ROWS, 3 FRAMES · NESTED-CARDS.HTML — RECURSIVE CARD SPLIT, ATOMIC IMG, 2 FRAMES · THREADWEBFLOW ×2 · RENDERWEBFLOW ×2

◪ *Both renders are scene layers (session state); the sources and their chains persist in metadata and container parts. The bake spread overleaf is where web content crosses into what survives the file being closed.* → *Appendix A*

The baked lane

The Year the Presses Ran Long

Twelve months, one flood, three second printings, and the first year since the rebuild in which **spoilage held under four percent**. The long account of how the counts were made.



January to March

The year opened the way press years usually do: cold plates, a cautious first run, and a circulation ledger that had been ruled but not yet written in. By the end of March the house had set **1,840 print copies** and logged 620 first opens of the digital edition — a ratio the board called healthy and the pressroom called a warning shot, and both were right.

What changed in the first quarter was not the volume but the discipline. The dock tally and the run counter were reconciled every week instead of every month, and the difference between them — the honest name for spoilage — stopped being a quarterly surprise and became a Tuesday routine.

> A copy is counted when it leaves the dock. Everything else this house believes follows from that sentence.

```
bakeWebFrame — this run's receipt:
native rectangles   2
native text frames  29
native paths        1
minted swatches     6 (Color/wb-…)
readout: (none published)
```

Everything the bake could not lower honestly it counted and reported instead — the deferred kinds above are the plugin keeping its books straight. What it did lower is ordinary page content now: these rectangles and text frames export to IDML and PDF like anything else in this annual, engine or no engine.

```
listPagedParts paged/media.paged.web/ → 0 part(s): none — every web source here fits its frame's label
```

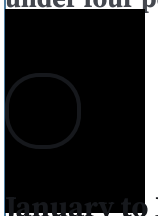

SPECIMEN NO. 174 · BAKEWEBFRAME — SWATCHES · RECTS · TEXT FRAMES · PATHS · DEFERRED KINDS COUNTED, PRINTED, UNCLAIMED · LISTPAGEDPARTS PAGED/MEDIA.PAGED.WEB/*

◪ Bake-to-native is partial (default-face text, solid rects, single-subpath paths), so this page demonstrates it and claims no row. It sits on a recto on purpose: on a verso the bake lands on the pasteboard. → Appendix A

The live lane

The Year the Presses Ran Long

Twelve months, one flood, three second printings, and the first year since the rebuild in which **spoilage held under four percent**. The long account of how the counts



January to March

The year opened the way press years usually do: cold plates, a cautious first run, and a circulation ledger that had been ruled but not yet written in. By the end of March the house had set **1,840 print copies** and logged 620 first opens of the digital edition — a ratio the board called healthy and the pressroom called a warning shot, and both were right.

What changed in the first quarter was not the volume but the discipline. The dock tally and the run counter were reconciled every week instead of every month, and the difference between them — the honest name for spoilage — stopped being a quarterly surprise and became a Tuesday routine.

> A copy is counted when it leaves the dock. Everything else this house believes follows from

This frame is being drawn by the web engine right now: a scene layer composed inside the frame, recomputed on every render. It is the better reading experience and the weaker artifact — session state, owned by the running plugin.

Turn back one leaf: the same source, flattened to native content that needs no engine at all.

SPECIMEN NO. 175 · SOURCE PANEL — MAKE WEB FRAME · EDITORS · SAVE TO DOCUMENT · RENDERWEBFRAME — THE LIVE SCENE LAYER

This page's render — when it paints — is a scene layer and will NOT survive the chapter checkpoint; the baked page overleaf will. That asymmetry is not a defect to apologise for; it is the difference between a view and an artifact. → Appendix A

18

The Catalog

A catalog is not a page decorated with numbers; it is a database wearing typography. Here a forty-eight-row order book registers with a query engine, its columns become bindings, and what lands on these pages — live fields, a native table, barcodes drawn as geometry — is resolved content, not pasted text.

The source, before any engine touches it — annual-orders.csv:

```
id,customer,region,product,qty,unit_price,order_date,sku,ean,upc,url,notes
1001,Bergfeld Booksellers,Alpine,Letterpress poster,120,4.80,2025-01-03,PA-…
1002,The Pass Hotel,Alpine,Seed catalogue,1000,5.60,2025-01-10,PA-2025-1002…
1003,Kettner Chocolatier,Alpine,Menu card,80,2.10,2025-01-17,PA-2025-1003-A…
1004,Quayside Fish Market,Harbour,Label sheet,220,0.85,2025-01-24,PA-2025-1…
1005,Harbour Light Theatre,Harbour,Shop ledger,60,12.50,2025-02-03,PA-2025-…
1006,Mole & Anchor Tavern,Harbour,Concert programme,90,1.35,2025-02-10,PA-2…
… 42 more rows
```

The record card

The order book from the opener, published. The CSV registers through the plugin's DuckDB query engine; the field--mapping wizard asks the Rust engine for each column's binding expression; and three of those bindings resolve into the card below as tagged placeholder fields, each placed at the text caret. Nothing in the card is typed: every value was resolved from record 1001 by the engine, and a data refresh re-resolves it in place.

ORDER RECORD — THREE LIVE FIELDS

Customer Bergfeld Booksellers

Product Letterpress poster

Unit price 4.84.8

The caret lane. Each field went in at a real insertion point: the page placed the caret after a label, and the plugin's lowering asked the host's caret door (C-9) where "here" was. A field is placed once — the session keys it by its binding id — so later lowering passes on these pages re-resolve these three in place rather than duplicating them.

All three fields above are live text variables the document keeps (paged:<binding>), not pasted text. The engine resolved them from the first order: customer → "Bergfeld Booksellers" · product → "Letterpress poster" · unit_price → "4.8". The expression behind each is the engine's own suggestion from the mapping wizard — a bare column reference in the plugin's publishing DSL.

SPECIMEN NO. 177 · IMPORTDATA · SHELL.PICKFILE · DUCKDB-WASM · MAP FIELDS… ×3 · LOWERALL → INSERTFIELD AT THE CARET · READ BACK VIA DOCUMENT.PLACEHOLDERS

Every new binding's field lands at the one caret, so this card took one wizard pass per line. The order book's date column is withheld: the query seam reads a DATE as epoch milliseconds where the engine expects days. → Appendix A

A native table, governed

The bound table is not a picture of a grid: the plugin inserts a frame, mints a real Paged table in its story, and pours every cell by table, row and column — one row per record of the order book. It reflows, it styles, and it exports to IDML as a table because it is one. Its columns are named in the bindings panel — order id, region, product and quantity — and each cell is resolved from the record, not typed.

id	region	product	qty
1001	Alpine	Letterpress poster	120
1002	Alpine	Seed catalogue	1000
1003	Alpine	Menu card	80
1004	Harbour	Label sheet	220
1005	Harbour	Shop ledger	60
1006	Harbour	Concert programme	90
1007	Plateau	Exhibition folio	150
1008	Plateau	Wedding invitation	250

The lowered table — minted with 49 native rows (one header, 48 records) over the four authored columns, trimmed here to eight records because a native table paints past its frame. Each column's header is its field name and each body cell the field's value for that record, as the engine resolved it.

The governance ledger — the dataset panel's own answers:

```
§7  11 columns · 0 documented · 11 schema notes
(a bare CSV travels with no metadata sidecar — every column arrives undocumented,
and the catalog says so)
§10 48 outputs from 48 records — 1001, 1002, 1003, 1004
(the panel pins the group key to the catalog's first column — id — so per-group
means per-record here; grouping by region is the engine lane the margin records)
§10 1 output from 48 records — catalog
§10 plan batch → disabled: "Needs a record-flow binding — add one in Bindings."
(this page authors no record-flow binding — the disabled button is the honest
answer)
§7.1 Provider "q_all-dataset" (revision 5f1373ebda6268e0) is shared with other
plugins.
```

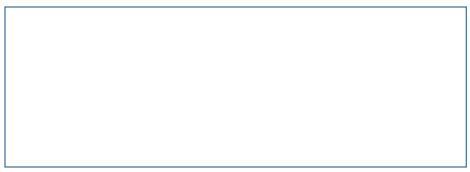

SPECIMEN NO. 178 · TABLE BINDING → INSERTTEXTFRAME + INSERTTABLE + CELL POUR · MOVED BY RESIZE-FRAME · GOVERNED CATALOG · BATCH PLANS · PUBLISH · REFUSAL

The grouped catalog this chapter wanted (per-region SUM and AVG footers) lives in the engine's record-flow lane; this page authors a flat table binding only, so it shows the native table and the refusals, verbatim. → Appendix A

Four symbologies, one order book

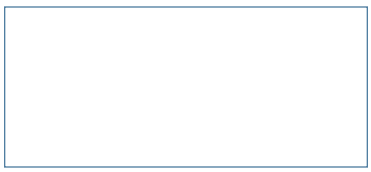
Retail wants EAN-13 and UPC-A, logistics wants Code-128, screens want QR — the order book carries a column for each, with real check digits. The engine encodes all four clean-room in Rust and lowers each symbol as native path geometry scaled to its bound frame: filled rectangles on the page, not an image in a frame. Print them at any size; they were never pixels.

Code-128 — sku



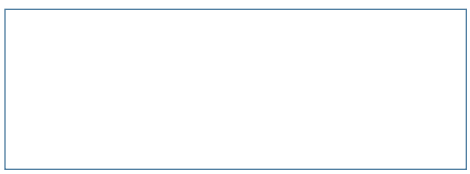
PA-2025-1001-ALP

EAN-13 — ean



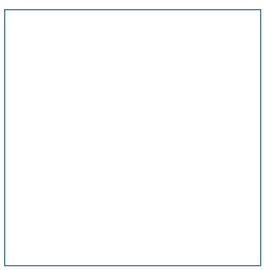
9007830010019

UPC-A — upc



(nothing drawn — see the run notes)

QR — url



https://paged.media/annual/orders/1001

3 of 4 symbologies drew — Code-128 58 · EAN-13 30 · QR 449 native filled paths, every module real page geometry. What did not draw refused honestly — the missing policy draws nothing rather than a fake symbol; the captions and the run notes carry each miss.

SPECIMEN NO. 179 · AUTHORING ROW ×4: CODE128/SKU · EAN13/EAN · UPCA/UPC · QR/URL · CLEAN-ROOM ENCODERS (RUST) → INSERTPATH FILLED RECTS · PAYLOADS PRINTED AS RESOLVED FOR RECORD 1001

◪ *A re-lower draws every bound symbol afresh instead of replacing it in place; this page removed the re-drawn copies of the expression page's two symbols. The raster barcode lane stays closed by design.* → *Appendix A*

Data sets and the refresh

A binding is live in both directions: the record card three pages back can be re-resolved against any record, and the values it shows at a moment can be captured as a named data set and applied back later — many fields, one undo step. The ledger below is this page's lab notebook: each line was read from the document's own field records at that step of the drive, and the card was left as the chapter found it.

Stepped to record 3 of 48 — the engine re-resolved every binding against that record and the card read customer = Kettner Chocolatier · unit_price = 2.1.

```
captured “Order 1001 — as imported” (record 1) — 1 variable(s)
captured “Order 1003 — Kettner” (record 3) — 2 variable(s)
before apply · customer = Bergfeld Booksellers · unit_price = 4.8
apply “Order 1003 — Kettner” → 3 written in ONE undo step · skips: none
after apply · customer = Kettner Chocolatier · unit_price = 2.1
refresh from sources → the fields KEPT the applied set (Overridden honored — a
refresh never clobbers what a designer pinned)
apply “Order 1001 — as imported” → 3 written · customer = Bergfeld Booksellers ·
unit_price = 4.8
```

Asked “what changed since last sync?” — the panel answered “changed since last sync: 0 changed · 0 unchanged · 3 added”: a first report is the baseline, so every binding arrives as added; the data itself has not moved.

Locale en → de left the bare-column price display unchanged (4.8) — a bare field reference renders the raw value; only the NUMBER/CURRENCY/PERCENT/DATEFMT format family consults the locale, and the expression page ahead drives that family. Measured, so said.

SPECIMEN NO. 180 · PREVIEW STEPPER (RESOLVE_AT) · CAPTUREDATASET / APPLYDATASET · RESOLVEBINDINGS + REFRESHFIELDS, MEASURED · CHANGE REPORT · LOCALE PROBE

◪ *The price-band rule this page planned (a data condition applying a cell style) is in the engine but has no editor surface: the bindings panel authors variable, image and barcode kinds only, so none is faked here. → Appendix A*

The expression language

paged.data speaks its own publishing DSL — not a spreadsheet grammar — and its whole vocabulary is registry-driven: the files printed below are the same ones the engine's build turns into its dispatch table, so a function missing here is uncallable by construction. Forty-two functions, five families, and below them two expressions evaluated live against the order book.

COLOR · 5
RGB · CMYK · HEX · SWATCH · ENUM

FORMAT · 4
NUMBER · CURRENCY · PERCENT · DATEFMT

LOGIC · 9
IF · AND · OR · NOT · ISBLANK · ISERROR · COALESCE · SWITCH · IFERROR

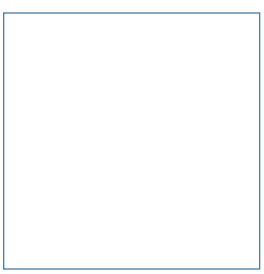
MATH · 11
ABS · ROUND · FLOOR · CEILING · MIN · MAX · SUM · MOD · POWER · TRUNC · SIGN

TEXT · 5
LEFT · RIGHT · SUBSTITUTE · MID · PROPER · FIND · TEXTJOIN · REPT

Two expressions, evaluated live and lowered as page geometry:



CURRENCY(unit_price) — the format family read record 1001's unit price and the engine encoded the formatted string as Code-128: 25 native filled paths, no image anywhere.



CONCAT(sku, " - ", url) — the text family joined the record's SKU and order URL and the result is this QR: 560 native filled paths.

SPECIMEN NO. 181 · 42 FUNCTIONS, 5 FAMILIES · BARCODE CURRENCY(UNIT_PRICE) · QR CONCAT(SKU, " - ", URL) · LOWERALL → INSERTPATH (VECTOR LANE)

The DuckDB network lane is dormant by consent design — remote sources stay inert until their origin is granted, and none is. No container part carries the imported data yet: the document travels with its metadata envelopes and the lowered native content, and a reopened session re-imports. → Appendix A

19

The Manuscript

A manuscript arrives as a .docx, and this chapter treats that arrival as a place, not an open: the Word file is poured into the layout as native content — real paragraphs and runs, styles synthesized from its own word/styles.xml — while the source bytes travel beside it as a container part. The spread that follows pours a full circulation report and reads it tier by tier; the closing page tells the save-back story straight.

A manuscript, poured

The circulation report below began as annual-report.docx. The plugin lowered its OOXML to native paragraphs and character runs, synthesized the styles its word/styles.xml declares, and poured it here — what you are reading is Paged content, not an embedded viewer, and it flows across this spread through an ordinary frame chain. The source file travels inside this container beside it.

Circulation Report

This is the year as the press room counted it: every copy that left the dock and every edition opened on a screen. Nothing is rounded upward.

The year in print

Print circulation rose in every quarter but the flooded third. The full-year total came to **8,315 copies**, up eleven percent on the year before. The digital editions grew faster still — 4,295 first opens against 2,700. In the house colour: **spoilage held under four percent** for the first time since the rebuild.

The fourth quarter's 2,390-copy run, the largest since the two-press years, went to bed on schedule.

How the counts are taken

Three old rules govern every number in this report:

- A copy is counted when it leaves the dock, not when it is printed.
  - Bundles broken for waste return to stock, uncounted.
  - Water-damaged bundles are waste, whatever the driver says.
- A digital edition is counted on its first open, once per subscriber.
- Nothing is ever counted twice, even when counting twice would look better.

Press operations

Each step of the calendar gates the next:

Tiers: the report's heading and caption styles were synthesized into this catalog and applied, its two list families carried over. Measured: they do not yet change the paint here.

Poured as one native story threaded across the spread. The outline panel's own summary of the retained lowering: 41 paragraphs · 1 table · 1 image · 2 links. The whole report paints within the chain.

SPECIMEN NO. 183 · PLACEDOC · SHELL.PICKFILE (PRE-ARMED CHOOSER) · OOXML → LOWEREDDOC → NATIVE POUR (SYNTHESIZED STYLES) · OUTLINE PANEL = THE RETAINED LOWERING'S RECEIPT

1. Plates are proofed against the signed-off layout.
    2. Registration is checked on the first fifty sheets.
    3. Colour bars are read, not eyeballed.
4. The run is released to the folder.
5. The dock tally is reconciled against the run counter.

Circulation by region

The Alpine routes remain the house's backbone.

Twelve months of counted copies		
Region	Channel	Copies
Alpine	Print	3,410
	Digital	1,240
Plateau	Print	2,000



The plate mark, as it appears on every signature this house folds.

Digital editions

The full tables live at paged.media/annual, the counting rules at docs.paged.media. Subscribers moved between the two channels: one readership, two doors.

Largest single run (Q4)..... 2,390 copies

Spoilage, full year.....3.7 percent

Editions gone to a second printing.....3

Next year: one more route, and spoilage below four. The presses are ready.

Tiers continuing: the table keeps its gridSpan title row and vMerge'd cell; the plate mark is an anchored picture whose link is a data URI carrying its own bytes; both Word hyperlink encodings lower to native links. Source part, from the container's own listing: paged/media.paged.doc/placed-mvh1hwizp8ouq8/source.docx

SPECIMEN NO. 184 · LINKFRAMES → ONE STORY ACROSS THE SPREAD · GRIDSPAN+VMERGE TABLE · PNG · BOTH HYPERLINK FORMS · LISTPAGEDPARTS → ONE SOURCE PART

◪ *The pour arrives in one frame sized to the .docx's own section margins; this spread re-framed and threaded it — ordinary layout, and the only kind of move there is.* → *Appendix A*

Save-back, honestly

The manuscript on the last spread can go home again. The exporter holds the report's original bytes, and when an edited story comes back it does not re-serialize the document — it patches it. quick-xml runs once as a locator, computing the exact byte ranges of the runs that changed; everything outside those ranges is raw-copied, and hand-built fragments are spliced into the holes. Untouched bytes are identical by construction — rsids, ignorable namespaces, parts Paged does not model — because no serializer ever touches them.

Which runs changed is decided by identity, not by index: the diff aligns story blocks, table rows and runs through an LCS keyed on what each node IS, so deleting a middle paragraph edits that paragraph's source node and never rewrites its successor. Synthesized docx-auto styles project back to direct formatting, so Word never meets a synthetic style name. And what the patcher will not risk, it refuses out loud: column operations on a gridSpan table land in a skip ledger instead of a guess.

This host's verdict, read from the plugin's own panel: LIVE — the structured story read (document.readStory@1) answers, so an edited story would flow back through the overlay, the diff and the byte-splice patcher.

Exported through the host exporter registry (the export center's row never rendered — see the run notes): annual-report.docx, 5,017 bytes, ZIP magic PK. The source was 4,978 bytes — the export differs, so the edited lane (or a re-zip) ran; recorded as measured.

Provenance, end to end: the report was PLACED, its source travels as the container part the last page printed, its content lowered to native stories — and the file that just left through the exporter is the same manuscript, patched or carried verbatim, never re-invented. A place, not an open; a patch, not a rewrite.

SPECIMEN NO. 185 · READINESS: LIVE (THE PANEL'S OWN VERDICT) · EXPORT CENTER → DOCX EXPORTER → DOWNLOAD · BYTE-SPLICE PATCHER, LCS IDENTITY DIFF

◪ *The report's two footnotes are not on these pages: the lowering reports them ("2 footnote(s) (numbering: Word's defaults: deci...") rather than fake a placement; they survive in every save-back.* *→ Appendix A*

Part III — The Press

SPECIMEN NO. 186 · D-PLATE — NO FURNITURE · AUTHORED FIELD + PART TITLE

Part I set the type and Part II ran the studios; Part III sends the book away. What follows is the press pass: the sheet's own anatomy drawn in the page's own geometry, a preflight the engine runs on itself, the container opened part by part, and the loss ledger — the exact list of what each exit can carry and what it must leave behind. Then the appendix, where every limit this book has recorded in its margins finally lands.

20 Imposition & Proof

The last chapter of the body is about leaving well: what the press needs, what the container carries, and what every exit admits it cannot take.

Six pages follow. A plate draws the sheet's anatomy — trim, bleed, marks — in the page's own native geometry. A preflight runs the real PDF export door against this very document and prints what it reported. The container is opened and its parts are listed, written and read back through the same doors a plugin uses. The loss ledger prints, verbatim, what the IDML interchange twin must leave behind. And the exit inventory names every artifact this book has already sent into other applications' hands.

Two doors are deliberately absent from those pages. paged.pdf's import opens a PDF as a new native document, and paged.publish's IDML open does the same for an interchange package — and both REPLACE the document they are invoked over. That is the correct shape for an import (a PDF becomes the document), and it is exactly why neither runs inline here: invoking either would end this book at this sentence. They are doors out of this build, exercised as exit paths at assembly — export, reopen, compare — and their import lanes are proven where imports are proven, by their own journeys. A demonstration that pretended otherwise would be a screenshot, not a specimen.

paged.pdf's import and publish's IDML open REPLACE the open document (host.nativeDocument.open, no dirty-state guard) - doors out of this build, exercised as exit paths at assembly, never simulated inline; their import lanes are proven by their own journeys. → Appendix A

trim rule

bleed · 9 pt

crop marks

registration

THE SHEET, ANNOTATED

Every page of this annual is a 540 × 720 pt sheet, and this one draws the press apparatus on itself: the Slate trim rule, the tinted nine-point bleed band outside it, crop marks held five points off the bleed, and the registration targets a press aligns its plates by — three of them, the fourth edge ceded to this book's own apparatus band below. Nothing here is decoration borrowed from a template — each mark is a native line, circle or rectangle this page minted over the wire.

And every one of them is a depiction. A page render stops at its own trim, so the anatomy is drawn 36 points in, where it can be seen. The real marks are minted outside the trim by the export door — exportPdf carries cropMarks, registrationMarks, colorBars and bleedOverridePt — in the assembly's PDF/X-4 press pass, on the finished book, where a proof belongs.

SPECIMEN NO. 188 · INSERTLINE ×22 · INSERTOVAL ×3 · DEPICTED — THE REAL MARKS LAND AT EXPORT

◪ *A page render stops at its trim, so this plate DEPICTS bleed and marks inside the sheet; the real ones are minted outside the trim by exportPdf (cropMarks / registrationMarks / bleedOverridePt) in the assembly's press pass.* → *Appendix A*

The preflight

A preflight is the press asking its questions early. While this page was built, the document you are holding went through the real export door — client.exportPdf, standard PDF 1.7, the same begin-page-finish session the File menu drives — and answered for every page it had. The serialized proof came back 15.1 MB long and was discarded on the spot; what this page keeps is the export's judgment: 1 diagnostic line(s) and 1 structured finding(s), printed verbatim below.

Two codes carry most preflights. font_not_embeddable names a face whose fsType licence forbids embedding: the exporter follows its policy — outline the glyphs, or fail the export — rather than ship a file the licence disallows. image_missing_bytes names a placed link whose pixels never arrived; the page paints its proxy, and a proof would too, so the finding says it before the press does. Each finding arrives with a severity and the page it was raised on, which is what turns a warning list into a work list.

Beneath the findings runs the flat diagnostics channel — 1 line(s) on this run, the first of them: “placed image could not be decoded; frame fill used instead”. The findings are the structured half of the same story (protocol 62 carries both), and this document elects to print the structured one.

Table 20·1 — preflight findings, exactly as the pdfExported reply carried them.

CODE	SEV.	PAGE	FINDING
image_decode_failed	warning	73	placed image could not be decoded; frame fill used instead

SPECIMEN NO. 189 · CLIENT.EXPORTPDF STANDARD=PDF17 (IN-MODULE) · BYTES DISCARDED · DIAGNOSTICS + FINDINGS KEPT

The preflight here runs standard pdf17 over the in-build document; the PDF/X-4 pass with output intent, ConvertToDestination and marks belongs to the assembly, on the finished book. → Appendix A

The container, opened

A .paged file is a valid IDML package with tenants. Every plugin that worked on this book may leave parts under its own paged/ namespace — a workbook, a web source, a recipe — and they travel with the file, opaque to the IDML half and untouched by it. Below is the container's own answer to what it carries at this moment of the build: 58 part(s), listed by the same listPagedParts door the editor's native-document backend reads.

Table 20·2 — the parts index, grouped by plugin. Roles are the bundles' own partTypes declarations; the path carries none.

PART	PLUGIN	ROLE
manifest-note.json	annual	note (this document)
model/blobs/25ab563c49cbb66e8eab126154859853	core	—
model/blobs/2a1356146e4a0f8f83185ec0ce77eb20	core	—
model/blobs/342d248ce00d821d2bad9cd4a3f544f1	core	—
model/blobs/42c2406dc5012029011c188ab46b3be8	core	—
model/blobs/5043d60d9be2a9a8dc40668fdee4636c	core	—
model/blobs/5282970ec234d6f4ee5fa4c1ea4381ad	core	—
model/blobs/58201b559eb48be0763b320712758df5	core	—
model/blobs/5fd42b18c0a74613f515f8c919d95d74	core	—
model/blobs/6124a34e012fcb7cf20e78f4376babb2	core	—
…	48 more part(s)	

ONE PART, WRITTEN AND READ BACK

```
writePagedPart paged/annual/manifest-note.json · caller unset · 153 bytes in
readPagedPart → found · 153 bytes out · note: “The full coverage ledger is written
at assembly.”
```

The write went through the same wire door a plugin's host.parts uses, with the caller field left unset — the engine then enforces only the paged/ boundary (a named caller is additionally confined to its own subtree). The bytes came back equal, and from the next save on, the note travels inside the file. It joins the 57 part(s) the studios had already checked in.

The loss ledger

Two exports leave this document. The .paged container keeps everything — the native model part carries the whole scene, masks and all. The .idml interchange twin is for other hands, and its writer holds a harder line: an unmutated document round-trips byte-identically (measured on the real corpus: 99 of 99 packages, 11,876 entries, zero gaps), a mutated one differs only where the edit touched — and whatever IDML cannot express is not smuggled or silently dropped, it is NAMED, in the export reply itself.



The specimen: a vermilion plate fading through an opacity mask whose artwork is the Annual Ramp. It stays on this page — paged-native, resident, and named by the export below as exactly what the interchange twin cannot carry.

Table 20·3 — idmlExported.lost, verbatim: every paged-native construct the IDML projection leaves behind.

No.	WHAT THE INTERCHANGE TWIN CANNOT CARRY
1	opacity mask on `ufe07b1` (artwork `ufe07b2`) is a paged-native construct — IDML has no opacity-mask element, so the mask is dropped and its artwork exports as an ordinary item. Save as `.paged` to keep it.
2	opacity mask on `ufe07b3` (artwork `ufe07b4`) is a paged-native construct — IDML has no opacity-mask element, so the mask is dropped and its artwork exports as an ordinary item. Save as `.paged` to keep it.
…	3 further entr(ies), same construct

Each entry names its target, its mask artwork and the lossless path (.paged). Among them is this page's own specimen: the ledger reports the rectangle beside it, specific and addressed.

SPECIMEN NO. 191 · EXPORTIDML + LOST (C-28, IN-MODULE) · 99/99 BYTE-IDENTITY CARRY-THROUGH CITED

Live paragraph-range overrides and full style definitions travel in document.pgm but are lost in the IDML projection - a recorded finding of this campaign; the container keeps them, the interchange twin does not. → *Appendix A*

The exit paths

A document that cannot leave is a terrarium. This book leaves by several doors, and the chapter closes by walking them in order of ceremony.

The press door is PDF/X-4. The export dialog's options ride the wire as one structure: standard pdfx4, an output-intent profile with its human-readable output condition, the colour policy — preserve numbers, or ConvertToDestination through the registered CMM — and the apparatus the marks plate depicted: cropMarks, registrationMarks, colorBars, pageInfo, and a bleed override in points. That pass is performed at assembly, on the finished book, where a proof belongs; this chapter ran the preflight for it three pages ago and printed what the door reported.

The interchange door is IDML. The same bytes InDesign opens, written by the carry-through writer whose discipline the loss ledger opposite just showed: byte-identical where nothing changed, honest about the little that cannot travel. And beneath both sits the container itself — the .paged file each chapter checkpoint already proves, which reopens with every plugin's parts intact.

Then the side doors, already used. The studios of Part II did not only import; they exported, and their artifacts left this build as real files: the darkroom's layered .psd, the ledger's .xlsx workbook re-encoded from the live grid, the manuscript's .docx save-back spliced byte-conservatively into the original, and the colour chapter's .ase swatch library, captured from the editor's own save door and re-imported to prove the loop. Each of those files opens in the application it was made for — that is what a sibling artifact is.

Two doors remain, and the honest line about them closed the opener: paged.pdf opens PDFs and paged.publish opens IDML, and both REPLACE the document they are invoked over. Their import lanes are proven by their own journey specs, not by this document — which prefers to remain open to its last page.

Appendix A — The Limits Ledger

A recorded limit is a demonstration that says where it stops. Every page of this annual that showed a feature partially, or met an edge the engine declares rather than hides, set a ◪ note in its margin and pointed here. This ledger is those margins, gathered: compiled at build time from the page sources themselves — every marginNote call read back out of the modules, deduplicated, its folio resolved through the chapter plan — because a limits appendix that was typed by hand would itself be a claim nobody checked.

The annual prints them for the same reason it prints specimen labels: a specimen that hides its edges is an advertisement. 76 limits stand recorded — 70 marked ◪, demonstrated to a recorded boundary, and 6 marked □, absences the platform declares by design rather than failures it met. Each entry is compacted to its leading clause; the full wording stands in the margin of the folio cited beside it.

□ No generate-TOC op exists on the wire; the fixture's TOC style is declared, the entries here are set live … · v

◪ Guides are an editor overlay: the wire holds them (SpreadSummary.guides) and the canvas draws them violet, … · 4

◪ The masterPages collection reports self ids only — label is a copy of the id — so the seven names above … · 6

◪ Where no Chapter Title paragraph precedes a page, the Running Header variable prints its baked ResultText — … · 8

◪ characterOtfFeatures stores a tag list the shaper never reads · 10

◪ The GSUB rows here demonstrate the authoring surface, not the substitution: the tag list persists, exports … · 11

◪ Optical kerning falls back to metrics — the middle kerning sample and the first render identically by … · 12

◪ H&J spacing limits and the hyphenation zone parse and render but have no property path · 15

◪ Keep flags ride the wire and persist, but the composer does not yet consult them when it fragments · 16

◪ The style-definition door carries 8 paths; font, leading and rules are basedOn or range matters · 19

◪ NextStyle fires on interactive typing, not on wire pours; the chains are shown as live-read data · 20

□ NumberingStartAt and per-level restarts are not on the wire; the lettered sub-list restarts by story scope · 21

◪ Paragraph shading and border parse + render but have no property path on the wire · 22

◪ A paragraph that crosses a frame boundary carries one line into the next frame and scatters further … · 24

◪ An oversized footnote does not split across frames · 25

◪ setFieldValue does not drive these two markers — they resolve at compose time · 26

◪ The wire sets the auto-sizing MODE only: AutoSizingReferencePoint — the anchor the growth holds — is parse- … · 28

□ A wrap contour excludes EVERY intersecting text frame — there is no per-frame ignore-text-wrap door — so … · 29

◪ The wire-minted anchored frame paints nothing, and property writes on it land in some runs and not in … · 30

◪ Two scripts, two shipped faces: Noto Sans Arabic for the Arabic and the mixed line, Open Sans for the … · 32
◪ Kinsoku is enforced as a hard rule keyed on the type's presence — the push-in/push-out flavours are … · 33
◪ Ruby renders as a centred group reading: a PerCharacter request collapses to the group form · 34
◪ insertHyperlink is the apparatus's one live door and it mints URL destinations only — the page-flavoured … · 35
◪ Bookmarks, cross-references and index markers are fixture-authored · 36
◪ frameTransform REPLACES the item transform (proved by the pair above) · 38, 39
◪ Polygon corners: only the TopLeft slot (or the global pair) drives geometry, uniformly across all corners · 38, 39
◪ Kind boundaries on the wire: frameStrokeEndCap and frameStrokeAlignment are rectangle-addressed (caps shown … · 40
◪ setStyleProperty answers UnsupportedProperty for the object collection (probed live) - an object style … · 42
□ IDML has no opacity-mask element. The .paged container keeps both pairs verbatim · 46
◪ The satin row's registry entry still records invert as ignored by the rasterizer (W1.8) · 50
◪ Overprint is honoured plane-aware on the CPU composite · 52
◪ The .ase format carries name, space, values and the spot flag — not tints and not alternates: Vermilion 20% … · 53
◪ Colour settings, proof setup, ink aliases and document defaults are app-level, non-undoable session state · 54
◪ cellFillTint is stored per cell: a read answers 30, and the second cell paints the vermilion at that tint … · 58
□ Nested tables render from IDML; the wire has no door (insertTable addresses a story, never a cell) · 59
◪ setStyleProperty refuses the cell and table style collections (no refusal recorded…), and decimal tab stops … · 60
◪ EPS is recognised, not decoded (no PostScript interpreter, by declaration): the frame renders its own fill … · 63
◪ Fitting enums, the reference point and auto-fit are persisted intent — the geometry above comes from the … · 64
◪ no pattern SWATCH exists to make (RFI C-31: no pattern paint type in IDML, the model or the wire) - the … · 65–76
◪ the importer's documented envelope is solid colours and geometry - gradients, <text> and <image> in an SVG … · 67
◪ The anchor editors compare a page-local pointer with spread-stored anchors, one page width apart on this … · 68
◪ the arrangement runs in raw path space and composes at most 12 inputs / 256 faces · 69
◪ Live Paint here is regenerable, not live: no persistent faces, gap tolerance or edge strokes, and a painted … · 70
◪ the front-most-layer bake is NOT a composite of N fills - it lowers the top opaque layer to the frame's … · 71
◪ Composing this page with the paint exhibits in place hangs the engine indefinitely against the grown … · 72
◪ updateRepeat's dissolve-and-regroup batch is refused by this engine · 73
◪ A blend refuses keys whose subpath structure differs, and text frames outright · 74
◪ corner presets are shown on RECTANGLES - the engine accepts the corner properties on more kinds but renders … · 75
◪ The trace is one-shot and fills-only (demonstrated, not claimed) · 76

◪ A kernel run is session state. Each tile here persists only because its adjusted pixels were exported and … · 78
◪ The selection itself is session state — a coverage field over ingested pixels, with no .paged … · 80
◪ Strokes live in the plugin's session and its own undo journal, not in the document's history · 81
◪ Paint and raster type are PARTIAL rows, demonstrated but unclaimed: strokes commit per gesture (Stage-B … · 82
◪ PSD structural edits are record edits on the retained parse: visible in the exported file, not on the … · 83
◪ Three stated limits: Apply's composite is session-only · 84
◪ The page lowering reads the workbook without its styles (getRangeLowered), so its cell fills and text … · 88
◪ Importing this workbook REPLACED the chart wall's workbook as the container part: workbook persistence is a … · 89
◪ The sheets engine has a pagination lane of its own (lowerPaginatedToChain) that no editor command reaches … · 90
◪ The sheetFromDataset command opens the datasets panel · 92
◪ The live render is a C-1 scene layer and will not survive this chapter's checkpoint reload · 94
◪ The flow renders above are scene layers — session state that the chapter checkpoint will drop · 95
◪ Both renders are scene layers (session state); the sources and their chains persist in metadata and … · 96
◪ Bake-to-native is partial (default-face text, solid rects, single-subpath paths), so this page demonstrates … · 97
◪ This page's render — when it paints — is a scene layer and will NOT survive the chapter checkpoint · 98
◪ Every new binding's field lands at the one caret, so this card took one wizard pass per line · 100
◪ The grouped catalog this chapter wanted (per-region SUM and AVG footers) lives in the engine's record-flow … · 101
◪ A re-lower draws every bound symbol afresh instead of replacing it in place · 102
◪ The price-band rule this page planned (a data condition applying a cell style) is in the engine but has no … · 103
◪ The DuckDB network lane is dormant by consent design — remote sources stay inert until their origin is … · 104
◪ The report is longer than this spread's two frames — its tail (the dot-leader ledger lines and the closing … · 106, 107
◪ The report's two footnotes are not on these pages: the lowering reports them ("…") rather than fake a … · 108
◪ paged.pdf's import and publish's IDML open REPLACE the open document (host.nativeDocument.open, no … · 111
◪ A page render stops at its trim, so this plate DEPICTS bleed and marks inside the sheet · 112
◪ The preflight here runs standard pdf17 over the in-build document · 113
◪ Live paragraph-range overrides and full style definitions travel in document.pgm but are lost in the IDML … · 115
□ The index markers are fixture-authored - no wire op mints a PageReference, so a live document can read and … · A·6

THE ENGINE FINDINGS

Ten of the notes above are surface readings of deeper seams. The campaign's own roll:

facing-spread coordinate seam — inserts re-base page-local anchors by the spread origin; transforms speak stored coords.
wasm 0.62 one-batch regroup regression — updateRepeat / redefineSymbol's dissolve-and-regroup batch is refused; green on protocol 60.
paragraph-granularity cross-frame flow — a split paragraph carries one line over and scatters the rest one line per frame.
no ignore-text-wrap door — a wrap contour excludes every intersecting frame; the self-seated pull quote is inexpressible.
position-keyed line-breaker sliver — certain exact rects compose character-wrapped while an identical rect a row down sets clean.
characterOtfFeatures is authoring-only — the tag list stores and round-trips; the shaper never reads it.
in-chain mutation cost scaling — ~3.8 s per op against the full book vs ~1 s solo; recompose scales with content.
DuckDB date-ingest refusal — the provider's serde refuses any result set carrying a DATE column, whole.
unstyled DOCX pour paint — synthesized styles apply over the story but the paint shows one face and size.
export-center missing plugin exporters — the plugin section never renders its rows; save-backs ride the exporter registry.

Appendix A — The Index

An index is apparatus that accumulated. Six slim tabs rode the fore-edge of verso pages through this book, and with the two apparatus exhibits of Chapter 7 they filed PageReference markers under the ten topics the document declares. Here the filing resolves: topic by topic, with the folios the markers earned.

Appendices 86
fore-edge tab, folio 86

Baskerville 24, 35
fore-edge tab, folio 24 · exhibit, folio 35

Colophon 36
exhibit, folio 36

Data tables 56
fore-edge tab, folio 56

Footnotes 68
fore-edge tab, folio 68

Grids 10
fore-edge tab, folio 10

Lab colour 38
fore-edge tab, folio 38

Spot colour 38
fore-edge tab, folio 38

Typography 10, 35
fore-edge tab, folio 10 · exhibit, folio 35

Vertical writing 68
fore-edge tab, folio 68

SPECIMEN NO. 194 · INDEXTOPICS LIVE · MARKERS RESOLVED FROM THE EXPORT · 10 TOPICS · 12 MARKER PAGES

HOW THE INDEX RESOLVED

Nothing opposite was typed from a plan. The topics are the live indexTopics collection; the folio lists come from the document itself — this module exported its own IDML, found every PageReference marker in the stories, walked each marker's story to its text frame and each frame to its page through the spreads' own transforms, and converted physical pages to folios by the sections math the front matter authored (body folio = physical − 10). The tab pages and the Chapter 7 exhibits are simply where the markers turned out to be.

The arithmetic, printed so it can be checked: 12 marker pages across 10 of the 10 declared topics, every one carried by a fore-edge tab or an apparatus exhibit. A topic no marker feeds would print an em dash — an index reports its markers, not its wishes.

◪ *The index markers are fixture-authored - no wire op mints a PageReference, so a live document can read and resolve its index apparatus but not grow it. Recorded here, in the ledger it belongs to.*

Colophon

This annual was set by the engine it describes. Its numbers below are read at build time from the campaign's own records — the per-chapter ledger fragments, the capability table, the introspect catalog, the container itself — and none of them is typed from memory.

Chapters built — 37, 100-front-matter through 311-repair-b, each reloading its predecessor's .paged checkpoint through the ordinary open door.

Registry rows claimed — 196, led by plugin-draw ×21 · frames-paths ×19 · typography ×17 · color-swatches ×11 · image ×11.

Wire mutation ops exercised — 117 of the capability table's 151.

Property paths written — 201 of the introspect catalog's 243.

Container parts — 58 riding in this document's paged/ namespace as it stands on this page.

The faces — seven families: Source Serif 4 · Fraunces · Space Grotesk · EB Garamond · JetBrains Mono · Noto Sans Arabic · Noto Sans JP.

The measure — 540 × 720 pt trim, facing pages; a six-column body grid with 12 pt gutters on a 432 pt measure, twelve columns on the data pages, and a 13 pt baseline rhythm every leading in the style battery is a multiple of.

The toolchain — the paged engine at wire protocol 74, @paged-media/canvas-wasm 0.74.0-dev, driven end to end through the editor's own mutation wire; no page in this book was drawn by anything else.

As of this chapter's build; the assembly ledger is definitive.

Set entirely by the engine it describes.